

🛡️ CYBERSECURITY ACADEMY MASTER ROADMAP v4.0 — ELITE EDITION

Version: 4.0 (Elite)

Student: Mohamed Mohamed

Duration: 72 Weeks (~18 Months)

Study Time: 20 hours/week (4 hrs/day × 5 days)

Total Hours: 1,440+

Philosophy: Offensive + Defensive + Automation + AI Security + Professionalism

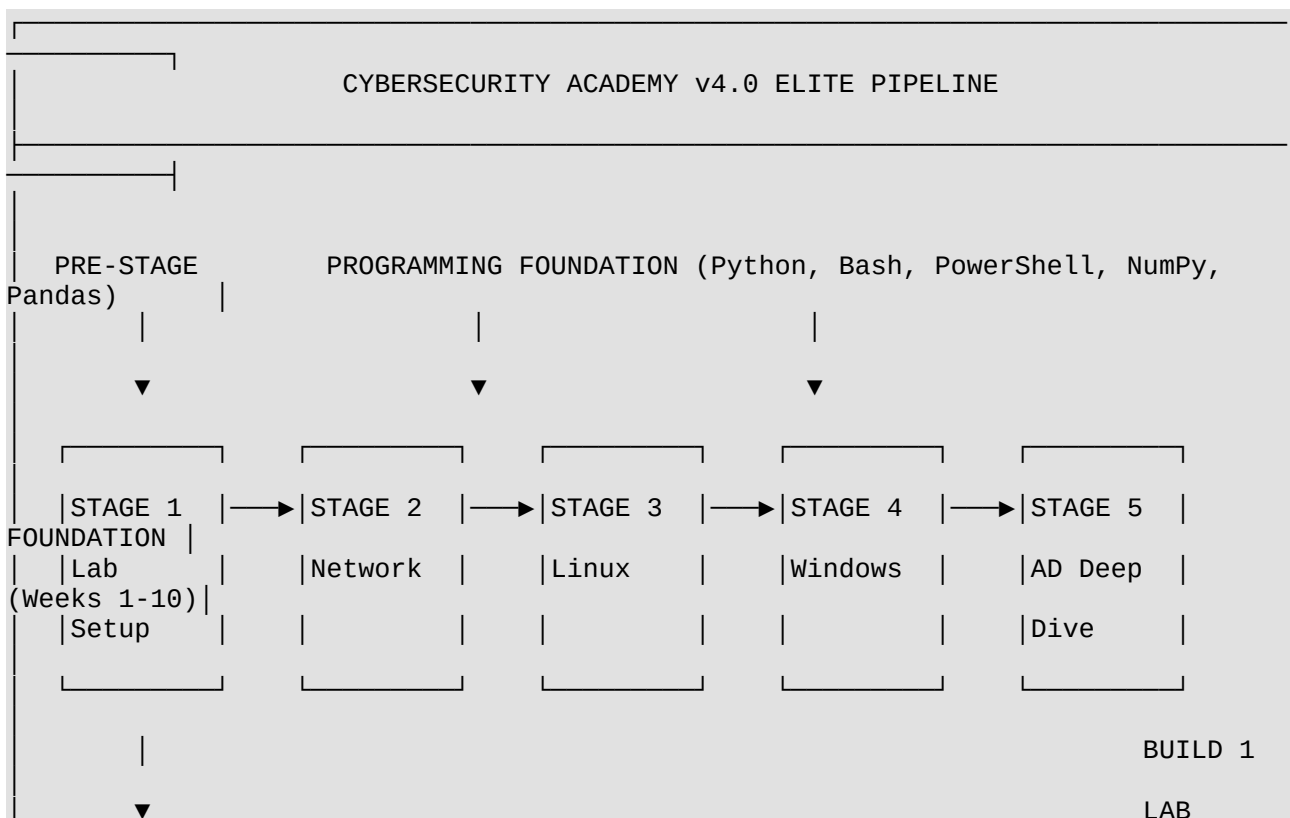
Output: Elite cybersecurity professional with GitHub portfolio, certifications, published research, and expert-level skills

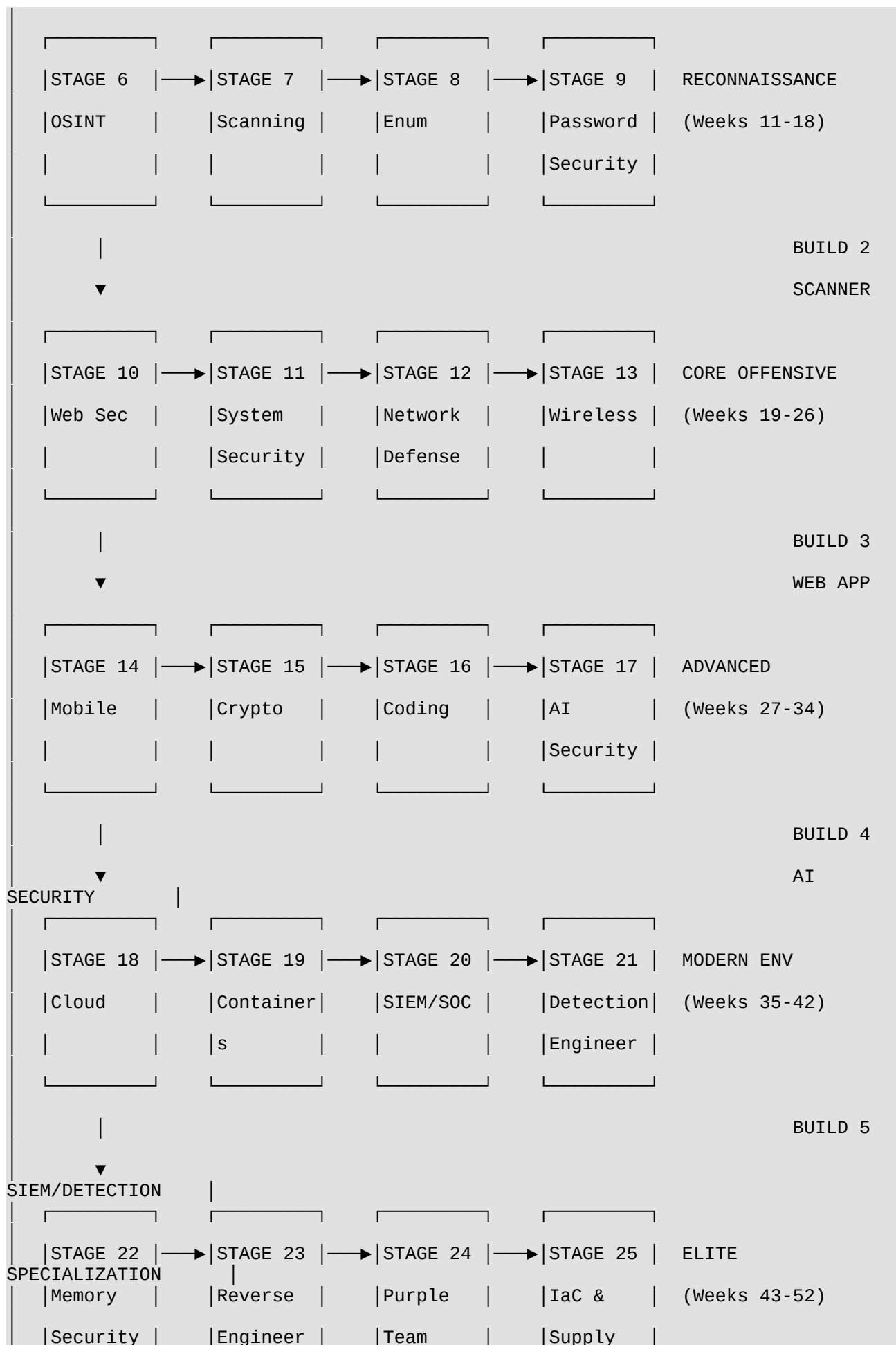
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ACADEMY STRUCTURE OVERVIEW {#academy-structure}





```

|           |   /Malware |           |   Chain   |
|_____|   _____|   _____|   _____|
|
|                                     BUILD 6
▼                                     ELITE
TOOLS |
|_____|   _____|   _____|
|STAGE 26|→|STAGE 27|→|STAGE 28| PROFESSIONALIZATION
|Pentest |   |Prof.  |   |Capstone| (Weeks 53-60)
|Method  |   |Skills |   |         |
|_____|   _____|   _____|

=====
| CERTIFICATIONS: Security+ → eJPT → PNPT → OSCP → OSWE/OSEP → AI Security
Certs |
| CTFs: PicoCTF → TryHackMe → Hack The Box → Pro Labs → Bug Bounty → DEF CON
CTF   |
| FREELANCE: Fiverr → Upwork → TCM Security → Independent → Elite Consulting
|
| JOBS: SOC Analyst → Pentester → Security Engineer → Red Team → AI Security
Lead |
|_____|

```

#	Topic	Description
P5	File I/O	Reading/writing files, CSV, JSON handling
P6	Error Handling	try/except, raise, finally, custom exceptions
P7	Bash Fundamentals	Shebang, variables, conditionals, loops, pipes
P8	PowerShell Basics	Cmdlets, variables, pipelines, objects
P9	Regex Basics	Pattern matching for log analysis and data extraction
P10	APIs & HTTP Requests	requests library, REST API interaction
P11	NumPy & Pandas Basics	Arrays, dataframes, data manipulation for ML
P12	Linear Algebra Review	Vectors, matrices, dot products (prep for AI)

Labs

1. Write a Python script that reads a log file and extracts IP addresses
2. Build a Bash script that checks system health (CPU, memory, disk)
3. Create a PowerShell script that lists all running services
4. Write a Python script that makes HTTP requests and parses JSON
5. Build a regex tool that finds email addresses in text files
6. Create a script that automates directory creation for academy structure
7. **Build a Python script that analyzes CSV data with Pandas**
8. **Implement basic matrix operations with NumPy**

Deliverables

- ☐ programming-foundation/ GitHub repo with 12+ scripts
- ☐ Personal script library (reusable functions)
- ☐ Cheat sheet: Python vs Bash vs PowerShell syntax comparison
- ☐ **NumPy/Pandas cheat sheet for security data analysis**

☐ **STAGE 1: FOUNDATION & LAB SETUP {#stage-1}**

Duration: 2 Weeks (Weeks 1-2)

Hours: 40

Purpose: Build the perfect practice environment

Topics

#	Topic	Hours	Description
1.1	Virtualization	6	VirtualBox, VMware, snapshots, networking modes
1.2	Kali Linux	8	Installation, configuration, tool categories, customization
1.3	Lab Network Design	4	NAT, bridged, host-only, internal networks, VLANs
1.4	VM Templates	4	Creating base images, cloning, linked clones
1.5	Vagrant & Automation	4	Infrastructure as code for labs, automated deployment
1.6	Docker Basics	4	Containers for isolated testing, Docker Compose
1.7	Documentation	4	Lab topology diagrams, IP schemas, password vaults
1.8	Backup Strategy	4	Snapshot management, VM exports, cloud backups
1.9	Troubleshooting	2	Common issues, network debugging, performance

Lab Environment

VM	OS	Purpose	IP
Kali	Debian-based	Attack platform	192.168.56.10
Metasploitable2	Ubuntu	Vulnerable Linux target	192.168.56.20
Metasploitable3	Windows	Vulnerable Windows target	192.168.56.21
Windows 10/11	Windows	Client testing	192.168.56.30
Windows Server	Windows Server	AD, services	192.168.56.31

VM	OS	Purpose	IP
Ubuntu Server	Ubuntu	Web services, databases	192.168.56.40
Windows Server 2019/2022	Windows Server	Active Directory lab	192.168.56.32
Elastic Stack	Ubuntu	SIEM/ELK stack	192.168.56.50
Splunk	Ubuntu	SIEM alternative	192.168.56.51
Docker Host	Ubuntu	Container security lab	192.168.56.60

Deliverables

- ☐ Complete virtual lab (9+ VMs operational)
- ☐ Network topology diagram (draw.io)
- ☐ Vagrantfile for automated lab deployment
- ☐ GitHub: /lab-setup/ with documentation
- ☐ **Lab access procedures and emergency recovery guide**

☐ STAGE 2: NETWORKING {#stage-2}

Duration: 2 Weeks (Weeks 3-4)

Hours: 40

Purpose: Master the foundation of all cybersecurity

Topics

#	Topic	Hours	Description	Attacker View	Defender View
2.1	OSI Model	3	7 layers, encapsulation, PDU, real-world mapping	Understand flow	Troubleshoot
2.2	TCP/IP	4	IPv4/IPv6, subnetting, CIDR, NAT, PAT	Target selection	Network design
2.3	Protocols	6	HTTP/HTTPS, DNS, DHCP, FTP, SSH, SMTP, SNMP	Protocol abuse	Protocol hardening
2.4	Routing	3	Static, dynamic (OSPF, BGP), routing tables	Route hijacking	Secure routing

#	Topic	Hours	Description	Attacker View	Defender View
2.5	Switching	3	VLANs, trunking, STP, port security	VLAN hopping	Segmentation
2.6	Wireshark	6	Capture, display filters, protocol analysis, expert info	Traffic analysis	Traffic monitoring
2.7	Packet Crafting	3	Scapy basics, custom packets, fuzzing	Custom attacks	Detection
2.8	Network Troubleshooting	4	ping, traceroute, netstat, ss, tcpdump, mtr	Find weak points	Maintain uptime
2.9	VPN & Tunneling	4	IPsec, SSL/TLS VPN, SSH tunneling, proxy chains	Bypass controls	Secure remote access
2.10	IPv6 Security	2	Dual-stack, tunneling, IPv6-specific attacks	IPv6 exploitation	IPv6 hardening

Labs (12 total)

1. Analyze TCP 3-way handshake in Wireshark
2. Analyze DNS query/response in Wireshark
3. Craft custom TCP packet with Scapy
4. Perform subnetting exercises (50+ problems)
5. Configure static routing between VMs
6. Configure VLANs on virtual switch
7. Capture and analyze HTTP vs HTTPS traffic
8. Perform ARP spoofing attack (lab only)
9. Set up SSH tunnel for traffic forwarding
10. Analyze DHCP handshake process
11. Perform network baseline and anomaly detection

12.Build network troubleshooting runbook

Deliverables

- ☐ **Network Analysis Report** (Wireshark captures + analysis)
- ☐ Subnetting cheat sheet (custom, with shortcuts)
- ☐ GitHub: /networking/ with captures and scripts
- ☐ Network topology diagram with security zones

Certifications After Stage 2

Certification	Provider	Why	Difficulty	Cost	Study Time
CompTIA Network+	CompTIA	Networking fundamentals (CRITICAL)	Medium	\$358	4-6 weeks
Cisco CCNA	Cisco	Deeper networking knowledge	Medium-Hard	\$300	8-12 weeks

☐ **STAGE 3: LINUX {#stage-3}**

Duration: 2 Weeks (Weeks 5-6)

Hours: 40

Purpose: Master the attacker's and defender's primary OS

Topics

#	Topic	Hours	Description	Attacker View	Defender View
3.1	Filesystem	3	Hierarchy, permissions, ACLs, special bits (SUID, SGID, sticky)	Privesc paths	Hardening
3.2	Users & Groups	3	/etc/passwd, /etc/shadow, sudo, PAM	Privesc targets	Access control
3.3	Processes	3	ps, top, htop, signals, cron, systemd	Process injection	Monitoring
3.4	Networking	4	netstat, ss, iptables, nftables, routing	Pivoting	Firewall rules

#	Topic	Hours	Description	Attacker View	Defender View
3.5	Shell Scripting	6	Bash advanced: functions, arrays, regex, error handling	Automation	Automation
3.6	Services	4	SSH, Apache, Nginx, MySQL, PostgreSQL	Service exploitation	Service hardening
3.7	Logging	3	rsyslog, journald, logrotate, remote logging	Log tampering	SIEM ingestion
3.8	Security Tools	4	fail2ban, AIDE, chkrootkit, Lynis, ClamAV	Bypass detection	Hardening
3.9	Package Management	2	apt, yum, dpkg, repositories, GPG keys	Supply chain	Integrity
3.10	Kernel & Modules	3	lsmod, modprobe, sysctl, kernel parameters	Rootkits	Kernel hardening
3.11	LVM & Storage	2	Logical volumes, encryption (LUKS), disk quotas	Data access	Data protection
3.12	Troubleshooting	4	Boot process, rescue mode, common issues	System knowledge	Recovery

Labs (14 total)

1. Configure SSH with key-based auth and disable password login
2. Set up iptables firewall with default-deny policy
3. Write Bash script that audits user permissions
4. Configure rsyslog for centralized logging
5. Set up fail2ban for SSH brute-force protection

6. Perform privilege escalation via SUID binary
7. Perform privilege escalation via sudo misconfiguration
8. Perform privilege escalation via PATH manipulation
9. Configure AIDE for file integrity monitoring
10. Set up LUKS disk encryption
11. Analyze Linux logs for intrusion detection
12. Build automated hardening script (CIS benchmarks)
13. Create custom systemd service
14. Document Linux hardening checklist

Deliverables

- ☐ **Linux Hardening Script** (CIS Level 1 + custom)
- ☐ **Privilege Escalation Playbook** (Linux)
- ☐ GitHub: /linux-security/ with scripts and documentation
- ☐ **Linux Security Audit Report**

Certifications After Stage 3

Certification	Provider	Why	Difficulty	Cost	Study Time
CompTIA Linux+	CompTIA	Linux administration	Medium	\$358	4-6 weeks
Red Hat RHCSA	Red Hat	Enterprise Linux	Medium-Hard	\$400	6-8 weeks

STAGE 4: WINDOWS {#stage-4}

Duration: 2 Weeks (Weeks 7-8)

Hours: 40

Purpose: Master the enterprise environment

Topics

#	Topic	Hours	Description	Attacker View	Defender View
4.1	Windows Architecture	4	NT kernel, subsystems, WOW64, registry, services	Understand targets	Architecture knowledge

#	Topic	Hours	Description	Attacker View	Defender View
4.2	File System	3	NTFS permissions, EFS, BitLocker, ADS	Data access	Data protection
4.3	Users & Groups	3	Local accounts, SAM, UAC, privileges	Privesc targets	Access control
4.4	PowerShell	8	Cmdlets, pipelines, remoting, scripting, modules	Lateral movement	Automation
4.5	Windows Services	3	SCM, service permissions, unquoted paths	Service abuse	Service hardening
4.6	Registry	3	hives, keys, values, persistence locations	Persistence	Monitoring
4.7	Event Logging	4	Event Viewer, wevtutil, forwarding, audit policies	Anti-forensics	Detection
4.8	Windows Firewall	2	Rules, profiles, IPSec, advanced security	Bypass	Rule design
4.9	Windows Defender	2	AV, ASR rules, exclusions, tamper protection	Evasion	Configuration
4.10	Sysinternals	4	Process Explorer, Autoruns, TCPView, ProcMon	System analysis	Troubleshooting
4.11	Windows Hardening	4	CIS benchmarks, LGPO, security baselines, STIGs	Bypass hardening	Implementation

Labs (14 total)

1. Analyze Windows registry for persistence mechanisms
2. Perform privilege escalation via unquoted service path
3. Perform privilege escalation via service permissions
4. Extract passwords from LSASS memory (lab only)
5. Configure Windows Event Log forwarding
6. Set up PowerShell remoting securely
7. Write PowerShell script for security audit
8. Configure Windows Defender ASR rules
9. Analyze Windows logs for intrusion indicators
10. Implement AppLocker application whitelisting
11. Configure BitLocker drive encryption
12. Build Windows hardening script (PowerShell)
13. Use Sysinternals to analyze running processes
14. Document Windows security baseline

Deliverables

- ☐ **Windows Hardening Script** (PowerShell, CIS-aligned)
- ☐ **Privilege Escalation Playbook** (Windows)
- ☐ GitHub: /windows-security/ with scripts and documentation
- ☐ **Windows Security Audit Report**

Certifications After Stage 4

Certification	Provider	Why	Difficulty	Cost	Study Time
CompTIA Security+	CompTIA	Security fundamentals (CRITICAL)	Medium	\$404	4-6 weeks
Microsoft AZ-900	Microsoft	Azure fundamentals	Easy	\$99	2-3 weeks
Microsoft SC-900	Microsoft	Security, compliance, identity	Easy	\$99	2-3 weeks

□ STAGE 5: ACTIVE DIRECTORY DEEP DIVE {#stage-5}

Duration: 2 Weeks (Weeks 9-10)

Hours: 40

Purpose: Master the backbone of enterprise environments

Topics

#	Topic	Hours	Description	Attacker View	Defender View
5.1	AD Architecture	6	Domains, forests, trees, trusts, sites, FSMO roles	Understand structure	Design
5.2	AD Objects	4	Users, groups, computers, OUs, GPOs, ACLs	Target selection	Organization
5.3	Authentication	6	Kerberos, NTLM, LDAP, certificate services	Protocol abuse	Protocol hardening
5.4	BloodHound	4	Enumeration, attack path analysis, custom queries	Path discovery	Path remediation
5.5	AD Attacks	8	Kerberoasting, AS-REP roasting, Golden Ticket, Silver Ticket, ACL abuse, DCSync	Domain compromise	Detection
5.6	Lateral Movement	4	Pass-the-hash, PSEXEC, WMI, WinRM, RDP, DCOM	Spread	Containment
5.7	AD Defense	4	Tiered administration, PAWs, LAPS, AD CS hardening, deception	—	Hardening
5.8	Azure AD	4	Cloud identity, hybrid identity, Conditional Access, PIM	Cloud attacks	Cloud defense

Labs (14 total)

1. Set up Active Directory lab domain
2. Configure AD users, groups, and OUs
3. Implement Group Policy Objects (GPOs)
4. Run BloodHound and analyze attack paths
5. Perform Kerberoasting attack
6. Perform AS-REP roasting attack
7. Perform Golden Ticket attack
8. Perform DCSync attack
9. Perform ACL abuse attack
10. Implement LAPS for local admin passwords
11. Configure tiered administration model
12. Set up Azure AD Connect
13. Configure Conditional Access policies
14. Build AD security assessment checklist

Deliverables

- ☐ **AD Attack Playbook** (10+ techniques with detection rules)
- ☐ **AD Security Assessment Report** on lab domain
- ☐ GitHub: /active-directory/ with BloodHound queries, PowerShell scripts, hardening GPOs
- ☐ **BloodHound Custom Queries Library** (20+ queries)

Certifications After Stage 5

Certification	Provider	Why	Difficulty	Cost	Study Time
CRTP	Pentester Academy	AD attacks practical	Hard	\$300	4-8 weeks
OSCP	Offensive Security	AD is exam topic	Hard	\$1,499+	3-6 months
PNPT	TCM Security	AD pentesting	Medium	\$300	4-6 weeks

STAGE 6: RECONNAISSANCE {#stage-6}

Duration: 2 Weeks (Weeks 11-12)

Hours: 40

Purpose: Learn what attackers know before they attack

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
6.1	Footprinting	4	Passive vs active, methodology, scope definition	Gather intel without detection	Understand exposure
6.2	OSINT	8	Framework, sources, tools, legal boundaries	Comprehensive target profiling	Assess public exposure
6.3	WHOIS	2	Domain registration, registrant info, privacy protection	Domain ownership discovery	Privacy protection
6.4	DNS Enumeration	6	Record types, zone transfers, brute force, DNSSEC	Infrastructure mapping	DNS hardening
6.5	Google Dorking	4	Operators (site, filetype, inurl, intitle), sensitive file discovery	Find exposed assets	Remove sensitive content
6.6	Metadata	2	EXIF, document properties, hidden data	Extract intelligence	Sanitize files
6.7	Shodan	4	Search engine for devices, filters, API	Find exposed IoT/industrial	Monitor exposure
6.8	TheHarvester	2	Email harvesting, subdomain discovery	Email enumeration	Email protection
6.9	Maltego	2	Visual link analysis, transforms	Relationship mapping	Asset inventory
6.10	Social Media OSINT	2	LinkedIn, Twitter, Facebook reconnaissance	Personnel targeting	Social media policy

#	Topic	Hours	Description	Attacker View	Defender View
6.11	Website Recon	2	Wappalyzer, builtwith, source code analysis	Technology stack discovery	Minimize exposure
6.12	Dark Web OSINT	2	Tor, dark web markets, leak sites, breach databases	Credential discovery	Breach monitoring
6.13	OSINT Automation	2	Python scripts for OSINT, APIs, data correlation	Scale intelligence	Monitor at scale

Labs (14 total)

1. Perform WHOIS lookup on target domain
2. Enumerate all DNS records (A, MX, NS, TXT, SOA, CNAME)
3. Attempt zone transfer (AXFR) on misconfigured DNS
4. Perform Google dorking for sensitive files (PDFs, docs, configs)
5. Search for exposed devices on Shodan
6. Extract metadata from PDFs and images
7. Run TheHarvester for email and subdomain enumeration
8. Create Maltego graph for target organization
9. Map social media presence of fictional target
10. Analyze website technology stack with Wappalyzer
11. Correlate findings from multiple OSINT sources
12. **Search dark web breach databases for leaked credentials**
13. **Write Python OSINT automation script**
14. Build OSINT methodology checklist

Deliverables

- ☐ **OSINT Report** on fictional target company (professional format)
- ☐ **Custom Google Dorks Cheat Sheet** (50+ dorks organized by category)
- ☐ GitHub: /osint/ with tools, outputs, and methodology
- ☐ **Reconnaissance Playbook** (Markdown, step-by-step)
- ☐ **Dark Web Monitoring Setup Guide**

□ STAGE 7: NETWORK SCANNING {#stage-7}

Duration: 2 Weeks (Weeks 13-14)

Hours: 40

Project: Python Network Scanner

Purpose: Map the attack surface systematically

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
7.1	Nmap Basics	6	Installation, syntax, host discovery, port states	Network mapping	Asset discovery
7.2	NSE (Nmap Scripting Engine)	6	Script categories, writing custom scripts, vuln detection	Automated exploitation	Vulnerability scanning
7.3	Port Scanning	4	TCP SYN, Connect, UDP, ACK, Window, Maimon scans	Stealth scanning	Detect scanning
7.4	Service Detection	3	Version detection (-sV), banner grabbing	Service fingerprinting	Service hiding
7.5	OS Detection	3	TCP/IP stack fingerprinting, -O flag	OS identification	OS hardening
7.6	Firewall Evasion	4	Fragmentation, decoys, spoofing, timing, source port	Bypass defenses	Detect evasion
7.7	Masscan	2	High-speed scanning, Internet-wide scanning	Large-scale recon	Rate limiting
7.8	Unicornscon	1	Alternative scanner, distributed scanning	Speed scanning	Detection
7.9	Scan Result	3	Output	Data	Vulnerability

#	Topic	Hours	Description	Attacker View	Defender View
	Analysis		formats, grepable, XML, parsing with Python	processing	triage
7.10	Python Network Programming	4	socket library, scapy, threading, async scanning	Custom tools	Custom monitoring
7.11	RustScan / Naabu	2	Modern fast scanners, integration with Nmap	Speed + accuracy	Detection tuning
7.12	Scan Detection & Defense	3	Detect port scans, IDS tuning, honeypot responses	Evasion	Detection

Labs (14 total)

1. Scan localhost with various Nmap flags
2. Perform TCP SYN scan vs Connect scan comparison
3. Perform comprehensive scan (-sV -sC -O -A)
4. Use NSE scripts for vulnerability detection
5. Perform UDP scan on common ports
6. Adjust timing templates (T0-T5) and analyze differences
7. Fragment packets to evade simple firewalls
8. Scan entire subnet and analyze results
9. Compare Nmap vs Masscan vs RustScan performance
10. Write Python script that parses Nmap XML output
11. Write custom Python port scanner with socket library
12. Write Python script that integrates Nmap and generates report
13. **Set up scan detection with Snort/Suricata**
14. **Build scan evasion vs detection lab**

Project: Python Network Scanner

Deliverables:

- ☐ `network_scanner.py` — Multi-threaded Python port scanner
- ☐ `nmap_parser.py` — Parse and analyze Nmap XML output
- ☐ `vuln_report.py` — Generate vulnerability report from scan data
- ☐ GitHub: `/network-scanning/` with scripts and documentation
- ☐ **Network Scanning Methodology Guide**
- ☐ **Scan Comparison Report** (different techniques analyzed)
- ☐ **Scan Detection Rules** (Snort/Suricata)

☐ **STAGE 8: ENUMERATION {#stage-8}**

Duration: 2 Weeks (Weeks 15-16)

Hours: 40

Purpose: Extract maximum information from discovered services

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
8.1	SMB Enumeration	4	enum4linux, smbclient, smbmap, shares, users, null sessions	Lateral movement	SMB hardening
8.2	FTP Enumeration	3	Anonymous access, banner grabbing, file listing	Data exfiltration	FTP hardening
8.3	SSH Enumeration	3	Version, algorithms, key types, banner	SSH exploitation	SSH hardening
8.4	SMTP Enumeration	3	VERFY, EXPN, RCPT TO, user enumeration	Email harvesting	SMTP hardening
8.5	LDAP Enumeration	3	Directory structure, users, groups, attributes	AD recon	LDAP hardening
8.6	SNMP Enumeration	3	Community strings, MIBs, snmpwalk, onesixtyone	Device info gathering	SNMP hardening
8.7	DNS	3	dnsenum,	Infrastructure	DNS

#	Topic	Hours	Description	Attacker View	Defender View
	Enumeration		dnsrecon, fierce, subdomain brute force	mapping	hardening
8.8	RPC & NFS Enumeration	2	rpcinfo, showmount, NFS shares	Unix lateral movement	NFS hardening
8.9	Database Enumeration	2	MySQL, PostgreSQL, MSSQL default configs	Database recon	Database hardening
8.10	Automated Enumeration	4	AutoRecon, Legion, custom scripts	Efficiency	Detection
8.11	Service-Specific Techniques	3	Telnet, RDP, VNC, HTTP methods	Service exploitation	Service hardening
8.12	Enumeration Countermeasures	3	Limiting information disclosure, monitoring	—	Defense
8.13	Kerberos Enumeration	2	User enumeration, AS-REQ analysis, SPN discovery	Pre-attack recon	Monitoring
8.14	Cloud Enumeration	2	AWS/Azure/GCP recon, IAM enumeration, storage discovery	Cloud recon	Cloud monitoring

Labs (14 total)

1. Enumerate SMB shares on Metasploitable2
2. Enumerate SMB users and permissions
3. Enumerate SNMP community strings
4. Enumerate LDAP directory structure
5. Enumerate FTP anonymous access and file listings
6. Enumerate SSH versions and supported algorithms
7. Enumerate SMTP users (VRFY, EXPN, RCPT)

8. Perform comprehensive DNS enumeration
9. Enumerate RPC services and NFS shares
10. Automate enumeration with AutoRecon
11. Correlate findings from multiple services
12. Build enumeration checklist by service type
13. Enumerate Kerberos users and SPNs
14. Enumerate AWS S3 buckets and IAM policies

Deliverables

- ☐ **Enumeration Cheat Sheet** (by service type: SMB, FTP, SSH, SMTP, LDAP, SNMP, DNS)
- ☐ **Automated Enumeration Script** (Bash or Python)
- ☐ GitHub: /enumeration/ with tools, outputs, and methodology
- ☐ **Service Enumeration Comparison Report**
- ☐ **Cloud Enumeration Playbook**

🛡️ STAGE 9: VULNERABILITY ASSESSMENT {#stage-9}

Duration: 2 Weeks (Weeks 17-18)

Hours: 40

Purpose: Systematically identify and assess vulnerabilities

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
9.1	Vulnerability Management Lifecycle	4	Discovery, assessment, prioritization, remediation, verification	Exploit window	Continuous security
9.2	CVE	2	Common Vulnerabilities and Exposures, numbering, searching	Find exploits	Patch management
9.3	CVSS Scoring	4	Base, temporal, environmental metrics, v3.1 calculation	Risk prioritization	Risk assessment

#	Topic	Hours	Description	Attacker View	Defender View
9.4	Nessus	6	Installation, configuration, scanning, policies, reporting	Scanner results	Enterprise scanning
9.5	OpenVAS	4	Open-source alternative, setup, scanning, reporting	Free scanning	Open-source VM
9.6	Nuclei	2	Fast vulnerability scanner, templates, community	Targeted scanning	Template-based detection
9.7	Manual Validation	4	Verifying scanner findings, false positive elimination	Accurate exploitation	Accurate assessment
9.8	Risk Analysis	3	Likelihood, impact, business context, risk matrix	Target selection	Resource allocation
9.9	Remediation Prioritization	3	Patch urgency, compensating controls, timelines	Patch gap exploitation	Efficient patching
9.10	Continuous Vulnerability Management	2	Scheduling, trending, metrics, reporting	Persistent access	Continuous improvement
9.11	Vulnerability Databases	2	NVD, Exploit-DB, VulDB, vendor advisories	Exploit research	Threat intelligence
9.12	Compliance Mapping	2	PCI-DSS, HIPAA, SOC 2, NIST vulnerability requirements	Compliance gaps	Compliance achievement
9.13	Cloud Vulnerability Scanning	2	ScoutSuite, Prowler, cloud-native	Cloud exploitation	Cloud assessment

#	Topic	Hours	Description	Attacker View	Defender View
			scanning		
9.14	Container Vulnerability Scanning	2	Trivy, Clair, Snyk container scanning	Container exploitation	Container security

Labs (14 total)

1. Research CVEs for discovered services from previous stages
2. Calculate CVSS scores manually for sample vulnerabilities
3. Install and configure Nessus (free version)
4. Run authenticated scan on Windows VM
5. Run unauthenticated scan on Metasploitable2
6. Install and configure OpenVAS
7. Compare Nessus vs OpenVAS results on same target
8. Validate scanner findings manually
9. Create risk matrix for all findings
10. Prioritize remediation by business risk
11. Document vulnerability lifecycle for one finding
12. Build vulnerability tracking spreadsheet/dashboard
13. Run ScoutSuite on AWS lab environment
14. Scan Docker images with Trivy and Snyk

Deliverables

- ☐ **Vulnerability Assessment Report** (professional format, 10+ pages)
 - ☐ **CVSS Scoring Guide/Calculator** (Python script or spreadsheet)
 - ☐ GitHub: /vulnerability-assessment/ with scans, analysis, and templates
 - ☐ **Scanner Comparison Report** (Nessus vs OpenVAS)
 - ☐ **Risk Matrix Template**
 - ☐ **Cloud & Container Vulnerability Report**
-

□ STAGE 10: AUTHENTICATION & PASSWORD SECURITY {#stage-10}

Duration: 2 Weeks (Weeks 19-20)

Hours: 40

Purpose: Understand the most common attack vector and how to defend it

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
10.1	Password Storage	3	Plain text, hashing, salting, peppering, key stretching	Identify weak storage	Secure storage design
10.2	Hashing Algorithms	4	MD5, SHA-1, SHA-256, SHA-3, bcrypt, Argon2, PBKDF2	Crack weak hashes	Select strong algorithms
10.3	MFA	4	Types (SMS, TOTP, hardware, biometrics), bypass techniques	MFA bypass	MFA implementation
10.4	Password Auditing	4	Tools, methodology, legal considerations, scope	Find weak passwords	Password policy enforcement
10.5	Authentication Attacks	6	Brute force, dictionary, spray, credential stuffing, pass-the-hash	Gain unauthorized access	Detect and block attacks
10.6	Password Policies	3	Length, complexity, rotation, history, lockout	Policy gaps	Strong policy design
10.7	Hashcat	6	Modes, rules, masks, wordlists, GPU acceleration	Hash cracking	Password strength testing
10.8	John the	4	Formats, rules,	Hash cracking	Password

#	Topic	Hours	Description	Attacker View	Defender View
	Ripper		incremental mode, custom wordlists		auditing
10.9	Rainbow Tables	2	Precomputed hashes, time-memory tradeoff, salt defense	Fast cracking	Salt implementation
10.10	Credential Dumping	3	Mimikatz, LSASS, SAM, NTDS.dit, cached credentials	Lateral movement	Credential protection
10.11	Kerberos Attacks	2	AS-REP Roasting, Kerberoasting, Golden Ticket	Domain compromise	Kerberos hardening
10.12	Cloud Identity Attacks	2	Cloud credential harvesting, IAM key abuse, SSRF to metadata	Cloud compromise	Cloud identity hardening
10.13	Passwordless Authentication	1	FIDO2, WebAuthn, passkeys, implementation	Bypass attempts	Modern auth design

Labs (14 total)

1. Generate hashes with MD5, SHA-256, bcrypt, Argon2
2. Crack MD5 hashes with Hashcat
3. Crack SHA-256 hashes with John the Ripper
4. Perform dictionary attack with custom wordlist
5. Perform brute force attack on short password (lab only)
6. Create and use rainbow tables (demonstrate salt defense)
7. Perform password spraying attack against test accounts
8. Bypass weak MFA implementations (lab only)
9. Analyze password strength of common patterns
10. Extract hashes from Windows SAM (lab VM)

- 11.Perform Kerberoasting attack (lab AD environment)
- 12.Test password policy effectiveness
- 13.Harvest cloud credentials via SSRF to metadata service
- 14.Implement FIDO2/WebAuthn demo

Deliverables

- ☐ Password Security Analysis Report (professional format)
 - ☐ Custom Wordlist Generator (Python script)
 - ☐ Hash Cracking Methodology Guide
 - ☐ GitHub: /password-security/ with tools, results, and documentation
 - ☐ Password Policy Recommendation Document
 - ☐ Cloud Identity Attack Playbook
-

☐ STAGE 11: WEB SECURITY {#stage-11}

Duration: 4 Weeks (Weeks 21-24)

Hours: 80

Project: Complete Web Application Security Assessment

Purpose: Master the most common attack surface

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
11.1	HTTP Deep Dive	4	Methods, headers, status codes, cookies, sessions, caching	Request manipulation	Request validation
11.2	Cookies	2	Attributes (Secure, HttpOnly, SameSite), session management	Session hijacking	Secure cookie config
11.3	Sessions	2	Session IDs, fixation, hijacking, timeout, regeneration	Session attacks	Secure session management
11.4	OWASP Top 10 (2021)	4	A01-A10 overview, risk,	Target selection	Defense prioritization

#	Topic	Hours	Description	Attacker View	Defender View
			examples		
11.5	XSS	6	Reflected, stored, DOM-based, CSP bypass, polyglots	Code injection	Input validation, CSP
11.6	CSRF	3	Token validation, double-submit, SameSite cookies	Unauthorized actions	Token implementation
11.7	SQL Injection	6	Error-based, Union, blind, time-based, stacked queries, WAF bypass	Data extraction	Parameterized queries
11.8	IDOR	3	Direct object references, predictable IDs, UUIDs	Unauthorized access	Access control
11.9	SSRF	3	Internal scanning, cloud metadata, protocol smuggling	Internal recon	URL validation
11.10	File Upload Security	3	Extension bypass, MIME type, magic bytes, content validation	Shell upload	Strict validation
11.11	Authentication Vulnerabilities	3	Brute force, credential stuffing, JWT attacks, OAuth flaws	Account takeover	Multi-layer auth
11.12	CORS	2	Origin validation, preflight, misconfigurations	Cross-origin attacks	Proper CORS config
11.13	Clickjacking	2	UI redressing, X-Frame-Options, CSP	UI manipulation	Frame protection

#	Topic	Hours	Description	Attacker View	Defender View
			frame-ancestors		
11.14	WebSockets	2	Protocol, security considerations, message manipulation	Real-time exploitation	WS validation
11.15	GraphQL Security	2	Introspection, query depth, batching, injection	API exploitation	Query limits
11.16	NoSQL Injection	2	MongoDB, Redis, CouchDB injection	Database exploitation	Input validation
11.17	API Security	3	REST, SOAP, GraphQL, rate limiting, authentication	API exploitation	API gateway security
11.18	Web Cache Poisoning	2	Cache keys, header manipulation, DOS	Cache exploitation	Cache validation
11.19	Race Conditions	2	Time-of-check time-of-use, limit overrun	Logic exploitation	Atomic operations
11.20	Prototype Pollution	2	JavaScript prototype chain, object injection	Logic exploitation	Object validation
11.21	LLM Web Security	2	Prompt injection, data exfiltration, jailbreaking	AI exploitation	Input filtering
11.22	Burp Suite	6	Proxy, repeater, intruder, scanner, extensions, sequencer	Web testing	Security testing
11.23	OWASP ZAP	3	Proxy, spider, scanner, fuzzer, API scanning	Free web testing	Security testing
11.24	Secure Coding	3	Input	—	Defense in

#	Topic	Hours	Description	Attacker View	Defender View
			validation, output encoding, parameterized queries, CSP		depth
11.25	Server-Side Template Injection (SSTI)	2	Jinja2, Twig, Smarty injection	RCE via templates	Template sandboxing
11.26	XXE & XML Attacks	2	XML external entities, billion laughs, quadratic blowup	Data exfiltration, DoS	XML parser hardening
11.27	Business Logic Flaws	2	Price manipulation, workflow bypass, access control flaws	Financial fraud	Logic validation

Labs (28 total)

1. Set up DVWA, WebGoat, and Juice Shop
2. Test for Reflected XSS with manual payload
3. Test for Stored XSS with persistent payload
4. Test for DOM-based XSS
5. Test for CSRF vulnerability
6. Test for IDOR by manipulating URLs/parameters
7. Perform error-based SQL injection
8. Perform Union-based SQL injection
9. Perform Boolean-based blind SQL injection
10. Perform Time-based blind SQL injection
11. Use SQLMap for automated exploitation
12. Test for file upload vulnerabilities (multiple bypasses)
13. Test for SSRF (internal scanning, metadata)
14. Test for authentication bypass techniques
15. Test for JWT vulnerabilities (none algorithm, weak secret)

16. Test for CORS misconfigurations
17. Test for clickjacking vulnerability
18. Analyze WebSocket security
19. Test GraphQL introspection and injection
20. Test for NoSQL injection
21. Test API endpoints for security flaws
22. Test for race conditions in limit logic
23. Test for prototype pollution
24. Document all findings with evidence and remediation
25. **Test for SSTI in multiple template engines**
26. **Test for XXE vulnerabilities**
27. **Test for business logic flaws in e-commerce apps**
28. **Perform full API security assessment**

Project: Complete Web Application Security Assessment

Deliverables:

- ☐ **Web Application Penetration Test Report** (professional, 20+ pages)
- ☐ **OWASP Top 10 Testing Checklist** (comprehensive, with test cases)
- ☐ **XSS/CSRF/IDOR/SSRF Demonstration** (screenshots + PoC code)
- ☐ **Secure Coding Examples** (PHP, Python, Node.js)
- ☐ GitHub: /web-security/ with test cases, PoCs, and fixes
- ☐ **Burp Suite/ZAP project file** with all saved requests
- ☐ **API Security Assessment Report**
- ☐ **LLM Security Testing Report**

STAGE 12: SYSTEM SECURITY {#stage-12}

Duration: 2 Weeks (Weeks 25-26)

Hours: 40

Purpose: Master post-exploitation and system-level attacks/defenses

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
12.1	Privilege	8	Linux: SUID,	Gain	Prevent

#	Topic	Hours	Description	Attacker View	Defender View
	Escalation		sudo, kernel exploits, path hijacking. Windows: services, registry, tokens, DLL hijacking	admin/root	escalation
12.2	Persistence Concepts	4	Startup items, registry, scheduled tasks, services, web shells, backdoors	Maintain access	Detect persistence
12.3	Scheduled Tasks	2	Cron (Linux), Task Scheduler (Windows), at	Persistence mechanism	Monitor scheduled tasks
12.4	Logging	4	Syslog, Windows Event Log, audit policies, log forwarding	Anti-forensics	Centralized logging
12.5	Defensive Hardening	6	CIS benchmarks, STIGs, minimal installation, least privilege, application whitelisting	Bypass hardening	Implement hardening
12.6	Lateral Movement	4	Pass-the-hash, PSEXEC, WMI, WinRM, SSH keys, RDP	Spread through network	Segment and monitor
12.7	Data Exfiltration	3	DNS tunneling, ICMP tunneling, HTTPS, cloud storage, steganography	Steal data	DLP, monitoring
12.8	Anti-Forensics	3	Log deletion, timestamp manipulation, rootkits, memory	Hide evidence	Immutable logs

#	Topic	Hours	Description	Attacker View	Defender View
			cleaning		
12.9	Incident Response Basics	3	Preparation, identification, containment, eradication, recovery, lessons learned	Understand response	Lead response
12.10	File Integrity Monitoring	2	AIDE, OSSEC, Tripwire, Windows FIM	Detect changes	Baseline monitoring
12.11	EDR Evasion	2	AMSI bypass, ETW bypass, syscall direct invocation	Evade detection	EDR tuning
12.12	Living Off The Land	2	LOLBAS, LOLBINS, native tool abuse	Stealth operations	Behavioral detection

Labs (14 total)

1. Identify SUID binaries for privilege escalation (Linux)
2. Exploit misconfigured sudo permissions (Linux)
3. Perform Windows privilege escalation via service misconfiguration
4. Perform Windows privilege escalation via unquoted service path
5. Perform token impersonation (Windows)
6. Create persistence via cron job (Linux)
7. Create persistence via registry run keys (Windows)
8. Create persistence via scheduled task
9. Analyze logs to detect privilege escalation attempts
10. Identify and remove persistence mechanisms
11. Perform lateral movement with PSEXEC
12. Perform lateral movement with SSH keys
13. **Implement EDR evasion techniques (lab only)**
14. **Build LOLBAS detection rules**

Deliverables

- ☐ **Privilege Escalation Playbook** (Linux & Windows, step-by-step)
 - ☐ **Persistence Detection Script** (Bash + PowerShell)
 - ☐ GitHub: /system-security/ with techniques and defenses
 - ☐ **Attacker vs Defender Perspective Report**
 - ☐ **Incident Response Runbook Template**
 - ☐ **EDR Evasion Lab Report**
-

♥ STAGE 13: NETWORK DEFENSE {#stage-13}

Duration: 2 Weeks (Weeks 27-28)

Hours: 40

Purpose: Build defensive infrastructure skills

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
13.1	Firewalls (Deep Dive)	6	Packet filter, stateful, application-layer, NGFW, WAF, rules, zones, NAT	Bypass techniques	Rule design, monitoring
13.2	IDS	6	Signature-based, anomaly-based, Snort, Suricata, rule writing	Evasion techniques	Detection engineering
13.3	IPS	4	Inline blocking, response actions, tuning, false positives	Bypass techniques	Response configuration
13.4	Network Segmentation	4	VLANs, DMZ, micro-segmentation, zero trust architecture	Lateral movement	Containment
13.5	Logging (Network)	4	NetFlow, sFlow, syslog, centralized logging, SIEM	Log deletion	Correlation, detection

#	Topic	Hours	Description	Attacker View	Defender View
			ingestion		
13.6	DDoS Mitigation	2	Attack types, mitigation strategies, rate limiting, CDN	DDoS execution	DDoS defense
13.7	Network Access Control	2	802.1X, MAC authentication, port security	Bypass	Access control
13.8	Honeypots	2	Low/high interaction, deployment, analysis	Detection evasion	Threat intelligence
13.9	Network Monitoring	4	Baseline, anomaly detection, behavioral analysis	Blend in	Detect anomalies
13.10	Defense in Depth	2	Layered security, compensating controls, redundancy	Find weakest link	Multiple layers
13.11	Zero Trust Architecture	2	Identity-based access, micro-segmentation, continuous validation	Identity compromise	Zero trust implementation
13.12	Threat Intelligence Platforms	2	MISP, OpenCTI, STIX/TAXII, threat feeds	Intel awareness	Threat intel integration

Labs (14 total)

1. Configure iptables rules on Linux (comprehensive rule set)
2. Configure Windows Defender Firewall with advanced rules
3. Install and configure Snort IDS
4. Create custom Snort rules for specific threats
5. Install and configure Suricata
6. Analyze IDS alerts and tune false positives
7. Design network segmentation scheme

8. Implement VLANs for security zones
9. Configure DMZ with pfSense
10. Set up centralized logging (rsyslog/syslog-ng)
11. Test firewall rule effectiveness with nmap
12. Attempt basic IDS evasion and analyze results
13. Deploy and monitor a honeypot
14. **Implement Zero Trust network policies**
15. **Set up MISP threat intelligence platform**

Deliverables

- ☐ **Firewall & IDS Configuration Guide** (comprehensive)
 - ☐ **Custom Snort/Suricata Ruleset** (10+ custom rules)
 - ☐ **GitHub: /network-defense/** with configs and documentation
 - ☐ **Network Segmentation Design Document**
 - ☐ **Defense Architecture Diagram**
 - ☐ **Zero Trust Implementation Guide**
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☐ **STAGE 14: WIRELESS SECURITY {#stage-14}**

Duration: 2 Weeks (Weeks 29-30)

Hours: 40

Purpose: Understand and secure wireless networks

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
14.1	Wi-Fi Fundamentals	4	802.11 standards, frequencies, channels, modes, frames	Understand targets	Design networks
14.2	WPA2	4	AES-CCMP, 4-way handshake, PSK vs Enterprise, KRACK	Handshake capture, cracking	Strong passwords, patches
14.3	WPA3	3	SAE, forward secrecy, PMF,	Limited attacks	Upgrade to WPA3

#	Topic	Hours	Description	Attacker View	Defender View
			transition mode		
14.4	Wireless Reconnaissance	4	airodump-ng, kismet, WiFi Analyzer, channel scanning	Target identification	Rogue AP detection
14.5	Deauthentication Attacks	2	Forced disconnect, handshake capture	Handshake capture	PMF, monitoring
14.6	WPA2 Cracking	4	aircrack-ng, hashcat, wordlist vs brute force	Password recovery	Strong passwords
14.7	Evil Twin Attacks	2	Rogue AP, captive portal, credential harvesting	Social engineering	Rogue AP detection
14.8	Bluetooth Security	2	Pairing, encryption, BLE, BlueBorne	Bluetooth attacks	Bluetooth hardening
14.9	Wireless Defense	4	Strong encryption, hidden SSID (limitations), MAC filtering (limitations), WPA3, monitoring	—	Comprehensive defense
14.10	Wireless Auditing	2	Site surveys, heat maps, coverage analysis, security audits	Find weak spots	Regular audits
14.11	5G Security	2	5G architecture, slicing security, edge computing risks	5G exploitation	5G hardening
14.12	IoT Wireless Protocols	2	Zigbee, Z-Wave, LoRaWAN	IoT exploitation	IoT hardening

#	Topic	Hours	Description	Attacker View	Defender View
			security		

Labs (14 total)

1. Set up wireless lab with USB adapter (Alfa AWUS036ACH or similar)
2. Capture wireless traffic with airodump-ng
3. Perform channel scanning and identify targets
4. Perform deauthentication attack (lab only)
5. Capture WPA2 handshake
6. Crack WPA2 PSK with aircrack-ng
7. Crack WPA2 PSK with Hashcat
8. Analyze wireless security configuration
9. Configure WPA3 on access point (if supported)
10. Implement wireless security best practices
11. Detect rogue access points
12. Document wireless audit methodology
13. **Analyze 5G security whitepaper and identify risks**
14. **Audit Zigbee/Z-Wave IoT device security**

Deliverables

- ☐ **Wireless Security Audit Report** (professional format)
- ☐ **Wireless Security Configuration Guide**
- ☐ GitHub: /wireless/ with captures and analysis
- ☐ **Wireless Attack Demonstration** (lab only, documented)
- ☐ **5G & IoT Wireless Security Assessment**

☐ **STAGE 15: MOBILE SECURITY {#stage-15}**

Duration: 2 Weeks (Weeks 31-32)

Hours: 40

Purpose: Understand mobile attack surface

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
15.1	Android Architecture	4	Linux kernel, libraries, Android Runtime, framework, apps	Understand targets	Secure architecture
15.2	APK Structure	3	Manifest, resources, classes.dex, assets, signature	Reverse engineering	App integrity
15.3	Mobile Threats	4	Malware, spyware, rooting, jailbreaking, sideloading, phishing	Attack vectors	Threat mitigation
15.4	Mobile App Security	4	Insecure storage, insecure communication, weak crypto, code injection	App exploitation	Secure development
15.5	Static Analysis	3	apktool, jadx, MobSF, decompiling, manifest analysis	Find vulnerabilities	Code review
15.6	Dynamic Analysis	3	Frida, Objection, runtime manipulation, traffic interception	Runtime exploitation	Runtime protection
15.7	iOS Security Basics	3	Architecture, sandbox, code signing, keychain, jailbreak detection	iOS exploitation	iOS hardening
15.8	OWASP Mobile Top 10	3	M1-M10, mobile-specific vulnerabilities	Target selection	Defense prioritization
15.9	Mobile	2	SSL pinning,	Traffic	Secure

#	Topic	Hours	Description	Attacker View	Defender View
	Network Security		certificate validation, MITM on mobile	interception	communication
15.10	Mobile Device Management	2	MDM, MAM, containerization, remote wipe	Bypass MDM	Enterprise mobile security
15.11	Flutter/React Native Security	2	Cross-platform framework security, code obfuscation	Framework exploitation	Secure development
15.12	Mobile API Security	2	Mobile-backend API testing, certificate pinning bypass	API exploitation	API hardening

Labs (14 total)

1. Set up Android emulator or physical device
2. Analyze APK with apktool and jadx
3. Review AndroidManifest.xml for permissions and components
4. Identify insecure data storage in app
5. Intercept mobile traffic with Burp Suite
6. Perform static analysis with MobSF
7. Perform dynamic analysis with Frida (intro)
8. Test for hardcoded credentials in APK
9. Test SSL pinning bypass
10. Analyze iOS app security (if Mac available)
11. Document mobile app security findings
12. Build mobile security checklist
13. **Analyze Flutter app for security flaws**
14. **Test mobile API endpoints for BOLA/BFLA**

Deliverables

- ☐ **Mobile Security Assessment Report** (professional format)

- ☐ **APK Analysis Template**
- ☐ GitHub: /mobile-security/ with tools and analysis
- ☐ **Mobile Security Testing Methodology**
- ☐ **Cross-Platform Mobile Security Guide**

☐ **STAGE 16: CRYPTOGRAPHY {#stage-16}**

Duration: 2 Weeks (Weeks 33-34)

Hours: 40

Purpose: Understand the math that protects everything

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
16.1	Encryption	6	Symmetric (AES, DES, 3DES), asymmetric (RSA, ECC), hybrid encryption	Key recovery, weak algorithms	Strong algorithm selection
16.2	Hashing	4	SHA family, MD5, BLAKE, collision resistance, HMAC	Hash collisions, length extension	Strong hashing
16.3	Digital Signatures	3	RSA signatures, DSA, ECDSA, signature verification	Signature forgery	Signature validation
16.4	PKI	4	Certificates, CAs, chains, revocation (CRL, OCSP), trust models	CA compromise, fake certs	PKI management
16.5	TLS/SSL	6	Handshake, cipher suites, versions, vulnerabilities (Heartbleed, POODLE, BEAST)	Downgrade attacks, weak ciphers	TLS hardening
16.6	Certificates	3	X.509, SAN,	Certificate	Certificate

#	Topic	Hours	Description	Attacker View	Defender View
			self-signed, Let's Encrypt, certificate pinning	spoofing	management
16.7	Cryptographic Attacks	4	Brute force, dictionary, rainbow tables, padding oracle, side-channel	Exploit weaknesses	Attack detection
16.8	Perfect Forward Secrecy	2	Ephemeral keys, session key compromise	Session decryption	PFS implementation
16.9	Key Management	2	Generation, storage, rotation, destruction, HSMs	Key extraction	Secure key lifecycle
16.10	Quantum Cryptography	2	Post-quantum algorithms, QKD, threat timeline	Future threats	Future-proofing
16.11	OpenSSL	3	Commands, certificate generation, testing, troubleshooting	Certificate analysis	Certificate management
16.12	Cryptographic Protocols	2	IPsec, SSH, S/MIME, PGP, secure messaging	Protocol exploitation	Protocol hardening
16.13	Homomorphic Encryption	2	Computing on encrypted data, Paillier, BFV/CKKS basics	Privacy attacks	Privacy-preserving ML
16.14	Zero-Knowledge Proofs	2	zk-SNARKs, zk-STARKs, applications	Proof forgery	Privacy authentication

Labs (16 total)

1. Generate RSA key pair with OpenSSL

2. Encrypt and decrypt files with AES-256-GCM
3. Encrypt and decrypt with RSA
4. Create and verify digital signatures
5. Analyze TLS handshake with Wireshark
6. Inspect certificate chain in browser
7. Test for weak cipher suites on web server
8. Implement HTTPS with Let's Encrypt
9. Demonstrate hash collision (MD5 vs SHA-256)
10. Analyze certificate transparency logs
11. Test for padding oracle vulnerability
12. Build cryptography toolkit script (Python)
13. Analyze SSH key algorithms
14. Document TLS hardening configuration
15. **Implement basic homomorphic encryption demo**
16. **Build zero-knowledge proof demonstration**

Deliverables

- ☐ **Cryptography Fundamentals Report** (comprehensive)
- ☐ **TLS Configuration Assessment Guide**
- ☐ GitHub: /cryptography/ with tools and examples
- ☐ **Certificate Analysis Script** (Python)
- ☐ **Cryptography Cheat Sheet** (algorithms, use cases, strengths/weaknesses)
- ☐ **Privacy-Preserving Cryptography Guide**

STAGE 17: SECURE CODING & AUTOMATION {#stage-17}

Duration: 2 Weeks (Weeks 35-36)

Hours: 40

Purpose: Build the automation muscle for security operations

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
17.1	Python for Cybersecurity	12	Requests, BeautifulSoup,	Tool development	Automation, detection

#	Topic	Hours	Description	Attacker View	Defender View
			Scapy, Impacket, Paramiko, Cryptography, Subprocess		
17.2	Bash Automation	6	Advanced scripting, one-liners, piping, process substitution, error handling	Quick exploits	System automation
17.3	PowerShell Automation	6	Advanced cmdlets, WMI, CIM, remoting, modules, error handling	AD attacks, lateral movement	Windows automation
17.4	Secure Coding Principles	8	Input validation, output encoding, parameterized queries, least privilege, fail securely, defense in depth	Find bypasses	Implement defenses
17.5	API Development Security	3	REST security, authentication, rate limiting, logging	API exploitation	Secure API design
17.6	CI/CD Security	3	Pipeline security, secret management, SAST, DAST, dependency scanning	Supply chain attacks	Secure pipelines
17.7	Infrastructure as Code Security	2	Terraform, CloudFormation, Ansible security	Misconfiguration	Secure IaC

Labs (14 total)

1. Write Python script that automates Nmap scanning and reporting

2. Write Python script that parses vulnerability scan output
3. Write Python script that automates web login brute force (lab only)
4. Write Bash script that automates system hardening
5. Write Bash script that monitors file integrity
6. Write PowerShell script that automates AD security audit
7. Write PowerShell script that collects Windows security events
8. Build Python tool that analyzes SSL/TLS configuration
9. Write secure Python web application with input validation
10. Write Python script that interacts with APIs securely
11. Build automation pipeline for security scanning
12. Write script that generates professional security reports
13. Create reusable security toolkit library
14. Document all tools with usage examples

Deliverables

- ☐ **Python Security Toolkit** (5+ reusable scripts)
- ☐ **Bash Automation Library** (5+ scripts)
- ☐ **PowerShell Security Toolkit** (5+ scripts)
- ☐ GitHub: /automation/ with complete toolkit
- ☐ **Secure Coding Guidelines Document**
- ☐ **Automation Pipeline Documentation**

☐ **STAGE 18: AI SECURITY & ADVERSARIAL ML {#stage-18}**

Duration: 2 Weeks (Weeks 37-38)

Hours: 40

Purpose: Master the intersection of AI and cybersecurity — critical for your Master's

Topics

#	Topic	Hours	Description	Attacker View	Defender View
18.1	ML Fundamentals for Security	6	Supervised/unsupervised learning, neural networks, training vs	Understand targets	Build detection

#	Topic	Hours	Description	Attacker View	Defender View
			inference		
18.2	Adversarial Examples	6	FGSM, PGD, Carlini & Wagner, evasion attacks	Bypass ML detection	Robust model training
18.3	Model Extraction & Inversion	4	Steal model weights, membership inference, model stealing	IP theft	Model protection
18.4	Data Poisoning & Backdoors	4	Training data manipulation, trigger-based backdoors	Corrupt models	Data integrity
18.5	LLM Security	6	Prompt injection, jailbreaking, RAG poisoning, data exfiltration	Exploit AI systems	Secure LLM deployment
18.6	AI Red Teaming	4	NIST AI RMF, OWASP ML Top 10, AI safety evaluations	Find AI vulnerabilities	AI security testing
18.7	Secure MLOps	4	Model signing, model cards, supply chain, monitoring	Supply chain attacks	Secure ML pipelines
18.8	AI for Defense	4	Anomaly detection, threat intelligence, automated response	—	AI-powered security
18.9	PyTorch/ TensorFlow Security	2	Framework vulnerabilities, serialization attacks	Exploit frameworks	Secure deployment

Labs (14 total)

1. Set up Python ML environment (PyTorch, TensorFlow, Jupyter)
2. Build and train a simple image classifier

3. Generate adversarial examples with FGSM against the classifier
4. Implement PGD attack and compare robustness
5. Perform membership inference attack
6. Poison training data and implant backdoor
7. Test prompt injection against LLM API
8. Perform jailbreaking on open-source LLM
9. Implement model extraction attack
10. Build AI-powered anomaly detector for network traffic
11. Secure ML pipeline with model signing
12. Test for LLM data exfiltration
13. Build AI red team assessment framework
14. Document AI threat model for enterprise system

Deliverables

- ☐ **AI Security Lab** GitHub repo with 4+ attack notebooks + defenses
- ☐ **AI Threat Model** for one real-world ML system
- ☐ **Adversarial Robustness Report** (compare model before/after hardening)
- ☐ **LLM Security Testing Report**
- ☐ **Secure MLOps Pipeline Documentation**

Certifications After Stage 18

Certification	Provider	Why	Difficulty	Cost	Study Time
None specific yet	—	Emerging field — build portfolio instead	—	Free	—
AWS ML Specialty	AWS	ML on AWS (security angle)	Hard	\$300	6-8 weeks
Google Professional ML Engineer	Google	ML engineering	Hard	\$200	6-8 weeks

STAGE 19: CLOUD SECURITY {#stage-19}

Duration: 2 Weeks (Weeks 39-40)

Hours: 40

Project: Secure a Cloud Environment

Purpose: Master the new default infrastructure

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
19.1	Cloud Fundamentals	4	IaaS, PaaS, SaaS, shared responsibility model, deployment models	Understand target	Design secure cloud
19.2	AWS Security	8	IAM, EC2, S3, VPC, CloudTrail, GuardDuty, KMS, Security Hub	AWS exploitation	AWS hardening
19.3	Azure Security	6	Azure AD, RBAC, NSG, Azure Security Center, Key Vault, Sentinel	Azure exploitation	Azure hardening
19.4	GCP Security	4	IAM, Compute Engine, Cloud Storage, VPC, Security Command Center	GCP exploitation	GCP hardening
19.5	IAM	6	Identity, policies, roles, permissions, federation, MFA, least privilege	Privilege escalation	Access control
19.6	Cloud Logging	4	CloudTrail, Azure Monitor, Cloud Logging, centralized logging	Log deletion	Monitoring
19.7	Cloud Misconfigurations	4	Public S3 buckets, open security groups, weak IAM, exposed databases	Find misconfigs	Prevent misconfigs

#	Topic	Hours	Description	Attacker View	Defender View
19.8	Cloud Pentesting	4	Scope, permissions, tools (ScoutSuite, Prowler, CloudSploit)	Cloud exploitation	Cloud assessment
19.9	Container Security in Cloud	2	ECS, EKS, AKS, GKE security	Container exploitation	Container hardening
19.10	Serverless Security	2	Lambda, Functions, Cloud Functions, injection, permissions	Serverless exploitation	Serverless hardening

Labs (14 total)

1. Set up AWS free tier account with MFA
2. Configure IAM users, groups, and policies with least privilege
3. Create and secure S3 bucket (public access blocked)
4. Set up VPC with proper security groups and NACLs
5. Audit AWS with ScoutSuite or Prowler
6. Analyze cloud misconfigurations
7. Set up Azure free account and configure Azure AD
8. Configure Azure NSG and RBAC
9. Set up GCP free account and configure IAM
10. Analyze GCP security configuration
11. Compare AWS/Azure/GCP security features
12. Test for S3 bucket misconfigurations
13. Test for IAM privilege escalation
14. Document cloud security assessment

Project: Secure a Cloud Environment

Deliverables:

- ☐ **Cloud Security Assessment Report** (multi-cloud)
- ☐ **IAM Policy Templates** (AWS, Azure, GCP)

- ☐ GitHub: /cloud-security/ with configs and documentation
- ☐ Cloud Security Checklist (multi-cloud)
- ☐ Cloud Hardening Scripts (Python/CLI)

Certifications After Stage 19

Certification	Provider	Why	Difficulty	Cost	Study Time
AWS Security Specialty	AWS	AWS security deep dive	Hard	\$300	6-8 weeks
Azure Security Engineer (AZ-500)	Microsoft	Azure security	Medium-Hard	\$165	4-6 weeks
GCP Security Engineer	Google	GCP security	Medium-Hard	\$200	4-6 weeks
CCSP	(ISC) ²	Cloud security comprehensive	Hard	\$599	2-3 months

☐ STAGE 20: CONTAINERS {#stage-20}

Duration: 2 Weeks (Weeks 41-42)

Hours: 40

Purpose: Secure the modern application packaging

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
20.1	Docker	6	Architecture, images, containers, Dockerfile, registry, networking	Container escape	Secure Docker
20.2	Docker Security	6	Image scanning, runtime security, seccomp, AppArmor, capabilities, user namespaces	Container escape	Container hardening
20.3	Kubernetes	6	Architecture,	K8s	K8s hardening

#	Topic	Hours	Description	Attacker View	Defender View
	Basics		Pods, services, deployments, configmaps, secrets	exploitation	
20.4	Kubernetes Security	6	RBAC, network policies, pod security, admission controllers, secrets management	K8s exploitation	K8s hardening
20.5	Container Orchestration Security	4	Helm security, service mesh (Istio), runtime security (Falco)	Supply chain attacks	Orchestration security
20.6	Supply Chain Security	4	Image signing, SBOM, dependency scanning, registry security	Supply chain poisoning	Supply chain integrity
20.7	Container Monitoring	4	Logging, metrics, tracing, Falco, Prometheus, Grafana	Evade detection	Container monitoring
20.8	Container Forensics	2	Container analysis, image forensics, runtime forensics	Anti-forensics	Incident investigation

Labs (14 total)

1. Build a secure Dockerfile with best practices
2. Scan Docker images for vulnerabilities (Trivy, Clair, Snyk)
3. Analyze container runtime security
4. Test for container escape vulnerabilities
5. Implement Docker secrets management
6. Configure Docker network security
7. Run Docker Bench for Security

8. Set up Minikube or kind cluster
9. Configure Kubernetes RBAC policies
10. Implement Kubernetes network policies
11. Apply Pod Security Standards
12. Enable and analyze Kubernetes audit logs
13. Scan cluster with kube-bench
14. Test for common Kubernetes misconfigurations

Deliverables

- ☐ **Secure Dockerfile Examples** (multiple scenarios)
- ☐ **Container Security Scanning Report**
- ☐ GitHub: /docker-security/ with Dockerfiles and configs
- ☐ **Kubernetes Security Configuration Guide**
- ☐ **RBAC and Network Policy Templates**
- ☐ GitHub: /kubernetes-security/ with manifests and docs
- ☐ **Container Threat Model Document**

Certifications After Stage 20

Certification	Provider	Why	Difficulty	Cost	Study Time
CKA	CNCF	Kubernetes administration	Hard	\$395	6-8 weeks
CKS	CNCF	Kubernetes security	Hard	\$395	4-6 weeks
Docker Certified Associate	Docker	Docker administration	Medium	\$195	4-6 weeks

STAGE 21: SIEM & SOC {#stage-21}

Duration: 2 Weeks (Weeks 43-44)

Hours: 40

Project: SOC Investigation

Purpose: Build defensive operations capability

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
21.1	Logging	4	Log types, formats, collection, forwarding, retention, integrity	Log deletion	Centralized logging
21.2	Splunk	8	SPL, search, dashboards, alerts, reports, apps, deployment	Evade detection	Detection engineering
21.3	ELK Stack	6	Elasticsearch, Logstash, Kibana, Beats, architecture	Evade detection	Detection engineering
21.4	Detection Rules	6	Sigma, YARA, Snort rules, correlation, threshold, anomaly	Evasion	Detection writing
21.5	Incident Investigation	6	Triage, scoping, timeline, evidence collection, containment	Anti-forensics	Investigation
21.6	SOC Structure	2	Tier 1/2/3, roles, responsibilities, shift work, metrics	Understand targets	Team design
21.7	SOAR	2	Automation, playbooks, orchestration, case management	Bypass automation	Automated response
21.8	Threat Intelligence	2	Feeds, IOCs, TTPs, ATT&CK, sharing	Understand defenses	Threat awareness
21.9	Digital Forensics Basics	2	Evidence handling, chain of custody, imaging,	Anti-forensics	Forensics

#	Topic	Hours	Description	Attacker View	Defender View
			analysis		
21.10	Metrics & KPIs	2	MTTD, MTTR, false positive rate, alert quality	—	Performance measurement

Labs (14 total)

1. Set up Splunk free trial or ELK stack
2. Ingest logs from multiple sources (Windows, Linux, firewall)
3. Create custom dashboards in Splunk/Kibana
4. Build correlation searches for brute force detection
5. Create alert for suspicious process execution
6. Create alert for lateral movement indicators
7. Investigate simulated security incident
8. Document incident timeline with evidence
9. Practice alert triage (10 alerts, prioritize)
10. Build SOC runbook for common incidents
11. Write Sigma rule for specific threat
12. Analyze SIEM use cases for effectiveness
13. Correlate logs across multiple sources
14. Build incident response checklist

Project: SOC Investigation

Deliverables:

- ☐ **SIEM Dashboard & Alert Configuration** (Splunk or ELK)
- ☐ **Incident Response Runbook** (3+ scenarios)
- ☐ GitHub: /siem-soc/ with configs and documentation
- ☐ **SOC Analyst Simulation Report**
- ☐ **Detection Rules** (Sigma or native format)
- ☐ **Incident Timeline Analysis**

Certifications After Stage 21

Certification	Provider	Why	Difficulty	Cost	Study Time
CompTIA CySA+	CompTIA	SOC analyst skills	Medium	\$404	4-6 weeks
Splunk Core Certified Power User	Splunk	Splunk proficiency	Medium	\$125	4-6 weeks
Blue Team Level 1	Security Blue Team	Blue team skills	Medium	\$200	4-6 weeks
GCIH	SANS/GIAC	Incident handling	Hard	\$7,000+	6+ months

☐ STAGE 22: DETECTION ENGINEERING & THREAT HUNTING {#stage-22}

Duration: 2 Weeks (Weeks 45-46)

Hours: 40

Purpose: Build proactive defense and detection capabilities

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
22.1	Detection Engineering Fundamentals	6	Detection as code, rule lifecycle, testing, coverage mapping	Bypass detection	Build robust detection
22.2	Sigma Rules	6	Syntax, conversion, generic signatures, field mapping, log sources	Rule evasion	Universal detection
22.3	YARA Rules	4	Pattern matching, memory scanning, malware detection, hunting	Evade signatures	Malware hunting
22.4	MITRE ATT&CK	4	TTP mapping, coverage gaps,	Understand coverage	Prioritize detection

#	Topic	Hours	Description	Attacker View	Defender View
	Mapping		detection validation, atomic tests		
22.5	Threat Hunting	6	Hypothesis-driven hunting, IOC vs IOA vs TTP, proactive searches	Evade hunters	Find advanced threats
22.6	Behavioral Analytics	4	Baseline, anomaly detection, UEBA, statistical analysis	Blend in	Detect anomalies
22.7	Jupyter for Security	4	Notebooks for log analysis, threat hunting, data visualization	—	Analysis platform
22.8	Detection Validation	4	Atomic Red Team, Caldera, purple team validation, coverage metrics	Test evasion	Validate effectiveness
22.9	False Positive Management	2	Tuning, baselining, threshold optimization, alert fatigue	Trigger false alerts	Efficient detection

Labs (14 total)

1. Write 5 Sigma rules for common attack techniques
2. Convert Sigma rules to Splunk and ELK queries
3. Write 5 YARA rules for malware families
4. Map detection rules to MITRE ATT&CK framework
5. Perform hypothesis-driven threat hunt on APT29 dataset
6. Build behavioral baseline for network traffic
7. Create anomaly detection with Python/Jupyter
8. Test detection rules with Atomic Red Team

9. Perform purple team exercise (attack + detection)
10. Analyze detection coverage gaps
11. Build detection engineering pipeline (CI/CD for rules)
12. Create threat hunting playbook
13. Validate detection effectiveness with red team
14. Document detection engineering methodology

Deliverables

- ☐ **Sigma Ruleset** (15+ rules mapped to MITRE ATT&CK)
- ☐ **YARA Ruleset** (10+ rules for malware detection)
- ☐ **Threat Hunt Report** (3 hypotheses, full timeline, IOCs)
- ☐ **GitHub: /detection-engineering/** with rules, hunts, documentation
- ☐ **Detection Coverage Matrix** (ATT&CK mapping)
- ☐ **Purple Team Exercise Report**

Certifications After Stage 22

Certification	Provider	Why	Difficulty	Cost	Study Time
GCIA	SANS/GIAC	IDS/IPS analysis	Hard	\$7,000+	6+ months
Blue Team Level 2	Security Blue Team	Advanced blue team	Hard	\$300	4-8 weeks
GCTI	SANS/GIAC	Threat intelligence	Hard	\$7,000+	6+ months

☐ **STAGE 23: MEMORY SECURITY & EXPLOIT DEVELOPMENT** **{#stage-23}**

Duration: 2 Weeks (Weeks 47-48)

Hours: 40

Purpose: Understand low-level exploitation

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
23.1	Memory Basics	4	RAM, virtual memory, paging,	Understand targets	Memory protection

#	Topic	Hours	Description	Attacker View	Defender View
			segmentation, memory layout		
23.2	Stack	4	Stack frame, ESP, EBP, EIP, local variables, parameters, return address	Stack overflow	Stack protection
23.3	Heap	3	Dynamic allocation, malloc, free, heap metadata, use-after-free	Heap exploitation	Heap protection
23.4	Registers	3	General purpose, segment, control, EIP, ESP, EBP, EAX	Register manipulation	Register monitoring
23.5	Buffer Overflow Theory	4	Overflow concept, EIP overwrite, shellcode, NOP sled	Theory understanding	Theory understanding
23.6	Practical Buffer Overflow	8	GDB, pattern_create, offset calculation, shellcode injection, bad characters	Exploit development	Detection
23.7	ASLR Bypass	2	Address Space Layout Randomization , information leaks, brute force	Bypass protection	Enable ASLR
23.8	DEP/NX Bypass	2	Data Execution Prevention, Return-Oriented Programming (ROP) intro	Bypass protection	Enable DEP
23.9	Format String	2	Format string vulnerabilities, read/write	Information leak, write	Input validation

#	Topic	Hours	Description	Attacker View	Defender View
			arbitrary memory		
23.10	Integer Overflow	2	Integer wrapping, signed/unsigned, size_t	Logic exploitation	Type checking
23.11	Use-After-Free	2	Dangling pointers, heap feng shui	Arbitrary code execution	Smart pointers
23.12	Modern Mitigations	2	Stack canaries, RELRO, PIE, FORTIFY_SOURCE, CFI	Bypass techniques	Defense layers
23.13	Windows Exploit Development	2	WinDbg, SEH, SEHOP, SafeSEH, structured exception handling	Windows exploitation	Windows hardening
23.14	Exploit Mitigation Bypasses	2	Modern bypass techniques, heap spray, info leaks	Advanced exploitation	Advanced defenses

Labs (14 total)

1. Analyze simple C program in memory with GDB
2. Examine stack layout with GDB
3. Identify vulnerable code patterns
4. Calculate buffer offset with pattern_create
5. Understand EIP overwrite concept
6. Analyze protection mechanisms on binaries (checksec)
7. Read and understand simple assembly
8. Trace function calls in GDB
9. Exploit simple stack buffer overflow
10. Generate and inject shellcode
11. Build exploit with Python/pwntools
12. Bypass simple stack protection (if possible)

13. Analyze and exploit format string bug
14. Practice on CTF-style challenges
15. **Build Windows buffer overflow exploit**
16. **Analyze modern mitigation bypass techniques**

Deliverables

- ☐ **Working Exploit PoC** (with full documentation)
- ☐ **Exploit Development Methodology Guide**
- ☐ GitHub: /exploit-dev/ with PoCs and writeups
- ☐ **CTF-Style Challenge Writeups** (3+)
- ☐ **Memory Architecture Explanation Document** (with diagrams)
- ☐ **GDB/WinDbg Debugging Cheat Sheet**
- ☐ **Windows Exploit Development Report**

Certifications After Stage 23

Certification	Provider	Why	Difficulty	Cost	Study Time
OSCP	Offensive Security	Buffer overflow is exam topic	Hard	\$1,499+	3-6 months
OSED	Offensive Security	Windows exploit development	Hard	\$1,499+	3-6 months
OSCE3	Offensive Security	Expert-level (OSCP+OSEP+OSED)	Expert	\$4,500+	6-12 months

☐ **STAGE 24: REVERSE ENGINEERING & MALWARE ANALYSIS {#stage-24}**

Duration: 2 Weeks (Weeks 49-50)

Hours: 40

Purpose: Master binary analysis and malware triage

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
24.1	Static Analysis	8	PE structure, ELF structure,	Hide functionality	Identify malware

#	Topic	Hours	Description	Attacker View	Defender View
			strings, imports/exports, packing, obfuscation		
24.2	Ghidra	6	Disassembly, decompilation, function analysis, cross-references, scripting	Understand code	Analyze threats
24.3	IDA Free / x64dbg	4	Disassembly, debugging, breakpoints, tracing	Dynamic analysis	Malware debugging
24.4	Dynamic Analysis	6	Sandboxing, API monitoring, behavior analysis, network traffic	Evade analysis	Understand behavior
24.5	YARA Rule Development	4	Pattern matching, memory scanning, hunting rules, sharing	Evade detection	Hunt malware
24.6	Malware Triage	4	Classification, IOC extraction, family identification, reporting	Hide identity	Rapid response
24.7	C2 Analysis	4	Protocol analysis, configuration extraction, domain extraction	Hide infrastructure	Disrupt operations
24.8	Anti-Analysis Techniques	4	Packing, VM detection, debugger detection, obfuscation	Evade analysis	Detect evasion

Labs (14 total)

1. Analyze PE structure with PE-bear and CFF Explorer

2. Perform static analysis on benign executable with Ghidra
3. Identify packed vs unpacked binaries
4. Unpack simple packed malware
5. Analyze strings and imports for IOCs
6. Perform dynamic analysis in sandbox (Any.Run, Cuckoo)
7. Debug malware with x64dbg (safe environment)
8. Extract C2 configuration from malware sample
9. Write YARA rule for malware family
10. Analyze ransomware behavior
11. Analyze info-stealer behavior
12. Analyze loader/dropper behavior
13. Build malware analysis report
14. Document malware triage methodology

Deliverables

- ☐ **Malware Analysis Report** (3 samples, professional format)
- ☐ **YARA Ruleset** (10+ custom rules)
- ☐ **GitHub: /malware-analysis/** with Ghidra projects, scripts, IOCs
- ☐ **Static Analysis Cheat Sheet** (Ghidra/IDA)
- ☐ **Dynamic Analysis Playbook**
- ☐ **Malware Triage Runbook**

Certifications After Stage 24

Certification	Provider	Why	Difficulty	Cost	Study Time
GREM	SANS/GIAC	Reverse engineering malware	Hard	\$7,000+	6+ months
GCFA	SANS/GIAC	Forensics analysis	Hard	\$7,000+	6+ months
OSCP	Offensive Security	Binary exploitation	Hard	\$1,499+	3-6 months

STAGE 25: PURPLE TEAM OPERATIONS {#stage-25}

Duration: 2 Weeks (Weeks 51-52)

Hours: 40

Purpose: Bridge offensive and defensive skills

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
25.1	Purple Team Concepts	4	Attack vs defense collaboration, continuous validation, metrics	Understand both sides	Optimize both sides
25.2	Atomic Red Team	6	MITRE ATT&CK atomic tests, execution, detection validation	Test techniques	Validate detection
25.3	CALDERA	4	Adversary emulation, automated attacks, plugin system	Emulate adversaries	Test defenses
25.4	Attack Simulation	6	Full kill chain simulation, APT emulation, scenario design	Realistic attacks	Realistic testing
25.5	Detection Validation	4	Alert verification, coverage gaps, false positive analysis	Bypass detection	Improve detection
25.6	Metrics & Reporting	4	Mean time to detect (MTTD), coverage percentage, improvement tracking	—	Measure effectiveness
25.7	Continuous Validation	4	Automated testing, BAS (Breach and Attack Simulation), purple team cycles	Persistent testing	Continuous improvement

#	Topic	Hours	Description	Attacker View	Defender View
25.8	Purple Team Leadership	4	Facilitation, communication, stakeholder management, program building	—	Build program
25.9	Adversary Emulation Frameworks	4	MITRE Caldera, Prelude Operator, Vectra, SCYTHE	Emulate threats	Test defenses

Labs (14 total)

1. Set up Atomic Red Team and execute 10 atomic tests
2. Map atomic tests to MITRE ATT&CK
3. Validate detection coverage for each test
4. Set up CALDERA and run adversary profile
5. Design custom adversary emulation scenario
6. Execute full kill chain simulation (recon → exfil)
7. Measure detection coverage before and after
8. Build detection for missed techniques
9. Perform purple team exercise with peer
10. Document attack vs detection mapping
11. Build purple team metrics dashboard
12. Create continuous validation pipeline
13. Present findings to fictional stakeholders
14. Build purple team program proposal

Deliverables

- ☐ **Purple Team Exercise Report** (3+ scenarios)
- ☐ **Atomic Red Team Test Library** (20+ tests)
- ☐ GitHub: /purple-team/ with scenarios, configs, results
- ☐ **Detection Coverage Dashboard**
- ☐ **Purple Team Metrics Report**

- ☐ Continuous Validation Pipeline

Certifications After Stage 25

Certification	Provider	Why	Difficulty	Cost	Study Time
GCTI	SANS/GIAC	Threat intelligence + emulation	Hard	\$7,000+	6+ months
GCIH	SANS/GIAC	Incident handling + purple team	Hard	\$7,000+	6+ months
OSCP	Offensive Security	Core offensive skills	Hard	\$1,499+	3-6 months

STAGE 26: INFRASTRUCTURE AS CODE & SUPPLY CHAIN SECURITY {#stage-26}

Duration: 2 Weeks (Weeks 53-54)

Hours: 40

Purpose: Secure modern DevOps and CI/CD pipelines

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
26.1	IaC Fundamentals	4	Terraform, CloudFormation, Ansible, Pulumi basics	Understand targets	Secure infrastructure
26.2	Terraform Security	6	State files, secrets management, module security, tfsec, checkov	Extract secrets	Secure configs
26.3	CI/CD Security	6	GitHub Actions, GitLab CI, Jenkins, pipeline security, secrets scanning	Supply chain attacks	Secure pipelines
26.4	Supply Chain Attacks	6	Dependency confusion,	Poison supply chain	Integrity verification

#	Topic	Hours	Description	Attacker View	Defender View
			typosquatting, SolarWinds, Codecov, 3CX		
26.5	SBOM & SLSA	4	Software Bill of Materials, SLSA framework, provenance, attestation	Hide malicious deps	Verify integrity
26.6	Container Supply Chain	4	Image signing, Cosign, Notary, registry security, base image scanning	Poison images	Verify images
26.7	Secret Management	4	Vault, AWS Secrets Manager, Azure Key Vault, GitHub secrets, rotation	Extract secrets	Secure secrets
26.8	DevSecOps Integration	4	Shift-left security, SAST, DAST, SCA, IaC scanning, policy as code	Bypass controls	Embed security
26.9	Git Security	2	Commit signing, pre-commit hooks, branch protection, audit	Tamper history	Verify commits

Labs (14 total)

1. Build secure Terraform configuration for AWS infrastructure
2. Scan Terraform with tfsec and checkov
3. Set up GitHub Actions pipeline with security scanning
4. Implement SAST (Semgrep, CodeQL) in CI/CD
5. Implement SCA (Snyk, Dependabot) in CI/CD

6. Build and sign container image with Cosign
7. Generate SBOM for application
8. Implement SLSA provenance attestation
9. Test for dependency confusion vulnerability
10. Set up HashiCorp Vault for secret management
11. Implement pre-commit hooks for secret scanning
12. Configure branch protection and commit signing
13. Perform supply chain security assessment
14. Build DevSecOps pipeline documentation

Deliverables

- ☐ **Secure IaC Templates** (Terraform, CloudFormation)
- ☐ **CI/CD Security Pipeline** (GitHub Actions/GitLab CI)
- ☐ **GitHub: /devsecops/** with configs and documentation
- ☐ **Supply Chain Security Assessment Report**
- ☐ **SBOM Generation and Verification Guide**
- ☐ **DevSecOps Implementation Playbook**

Certifications After Stage 26

Certification	Provider	Why	Difficulty	Cost	Study Time
HashiCorp Terraform Associate	HashiCorp	IaC fundamentals	Medium	\$70	2-4 weeks
AWS DevOps Engineer	AWS	CI/CD on AWS	Hard	\$300	6-8 weeks
CSSLP	(ISC) ²	Secure software lifecycle	Hard	\$749	3-6 months

☐ **STAGE 27: PENETRATION TESTING METHODOLOGY** **{#stage-27}**

Duration: 2 Weeks (Weeks 55-56)

Hours: 40

Purpose: Professionalize offensive skills

Detailed Topics

#	Topic	Hours	Description	Attacker View	Defender View
27.1	Methodology	4	PTES, OWASP Testing Guide, NIST 800-115, custom frameworks	Structured approach	Understand testing
27.2	Scoping	3	Rules of Engagement, IP ranges, exclusions, testing windows, legal	Scope understanding	Scope definition
27.3	Recon	4	Passive, active, OSINT, validation	Intelligence gathering	Exposure assessment
27.4	Scanning	3	Network, service, vulnerability scanning	Surface mapping	Asset discovery
27.5	Exploitation (Authorized Lab)	6	Gaining access, validation, impact assessment, evidence collection	Controlled exploitation	Understand impact
27.6	Post-Exploitation	4	Privilege escalation, persistence, lateral movement, data access	Impact demonstration	Understand risk
27.7	Reporting	6	Executive summary, technical findings, risk ratings, evidence, remediation	Client communication	Actionable intelligence
27.8	Client Communication	3	Status updates, findings briefings, remediation support	Relationship management	Vendor management

#	Topic	Hours	Description	Attacker View	Defender View
27.9	Tools of the Trade	3	Burp Suite Pro, Cobalt Strike, Metasploit Pro, custom tools	Professional tools	Detection awareness
27.10	Specializations	4	Web, network, mobile, wireless, social engineering, physical	Niche expertise	Niche defense

Labs (12 total)

1. Draft a penetration testing proposal
2. Create Rules of Engagement document
3. Perform scoped reconnaissance on lab environment
4. Map full attack surface
5. Prioritize vulnerabilities by risk and impact
6. Simulate full penetration test on lab
7. Document exploitation with evidence
8. Practice post-exploitation techniques
9. Draft initial findings report
10. Create executive summary
11. Practice client presentation
12. Review and refine methodology

Deliverables

- ☐ **Penetration Testing Methodology Guide** (comprehensive)
- ☐ **Sample RoE and Proposal Templates**
- ☐ GitHub: /pentest-methodology/ with templates
- ☐ **Methodology Comparison** (PTES vs OWASP vs NIST)
- ☐ **Sample Penetration Test Report** (full format)

Certifications After Stage 27

Certification	Provider	Why	Difficulty	Cost	Study Time
OSCP	Offensive Security	Industry standard	Hard	\$1,499+	3-6 months

Certification	Provider	Why	Difficulty	Cost	Study Time
		pentest			
PNPT	TCM Security	Practical network pentest	Medium	\$300	4-6 weeks
eCPPT	eLearnSecurity	Practical pentesting	Medium	\$400	4-6 weeks
GPEN	SANS/GIAC	Advanced pentesting	Hard	\$7,000+	6+ months
OSWE	Offensive Security	Web app pentesting	Hard	\$1,499+	3-6 months
OSEP	Offensive Security	Advanced pentesting	Hard	\$1,499+	3-6 months

□ STAGE 28: PROFESSIONAL SKILLS {#stage-28}

Duration: 2 Weeks (Weeks 57-58)

Hours: 40

Purpose: Transform technical skills into career success

Detailed Topics

#	Topic	Hours	Description	Purpose
28.1	Professional Report Writing	6	Executive summary, technical findings, risk ratings, evidence, recommendations, formatting	Client communication
28.2	Git & GitHub (Advanced)	4	Branching, merging, pull requests, actions, pages, portfolio organization	Version control
28.3	Documentation	4	Markdown, technical writing, diagrams, runbooks, SOPs	Knowledge sharing
28.4	Portfolio Website	4	GitHub Pages, Jekyll, Hugo, personal branding, projects showcase	Professional presence
28.5	Resume	3	Cybersecurity-	Job applications

#	Topic	Hours	Description	Purpose
			focused, ATS-friendly, achievements, metrics	
28.6	LinkedIn	3	Profile optimization, networking, content creation, personal brand	Professional network
28.7	Interview Preparation	6	Technical questions, behavioral (STAR), mock interviews, salary negotiation	Job success
28.8	Freelancing	3	Platforms, pricing, contracts, client management, taxes	Independent work
28.9	Career Planning	3	Paths, specializations, certifications, continuous learning, mentorship	Long-term growth
28.10	Soft Skills	3	Communication, presentation, teamwork, time management, ethics	Professional success
28.11	Research & Publication	2	Writing blog posts, CTF writeups, CVE submissions, conference talks	Thought leadership
28.12	Open Source Contribution	2	Contributing to security tools, bug fixes, feature development	Community reputation

Labs (14 total)

1. Write executive summary for sample findings
2. Document technical findings with evidence and screenshots
3. Create risk matrix visualization

4. Build professional report template (Word/LaTeX)
5. Practice client presentation (record yourself)
6. Peer review sample reports (good vs bad)
7. Create vulnerability evidence screenshots
8. Build remediation priority matrix
9. Optimize LinkedIn profile with keywords
10. Draft cybersecurity-focused resume
11. Practice technical interview (20 questions)
12. Practice behavioral interview (STAR method)
13. **Write and publish first blog post/CTF writeup**
14. **Contribute to open-source security tool**

Deliverables

- ☐ **Professional Penetration Testing Report Template**
- ☐ **Sample Executive Summaries** (3+)
- ☐ **GitHub: /report-writing/** with templates and examples
- ☐ **Report Writing Style Guide**
- ☐ **Portfolio Website** (GitHub Pages, live)
- ☐ **Optimized Resume** (cybersecurity-focused)
- ☐ **LinkedIn Profile** (optimized, active)
- ☐ **Interview Preparation Document** (50+ questions)
- ☐ **Freelancing Rate Sheet**
- ☐ **Career Roadmap Document**
- ☐ **Published Blog Post or CTF Writeup**
- ☐ **Open Source Contribution**



STAGE 29: PRACTICE & CAPSTONE {#stage-29}

Duration: 6 Weeks (Weeks 59-64)

Hours: 120

Purpose: Cement skills and prove competency

Detailed Topics

#	Topic	Hours	Description	Purpose
29.1	CTF Methodology	4	Approach, time management, tool selection, note-taking, teamwork	Competition success
29.2	Beginner CTFs	20	PicoCTF, TryHackMe, OverTheWire, CyberDefenders, BTLO	Skill building
29.3	Intermediate CTFs	20	Hack The Box, Proving Grounds, Root-Me, VulnHub	Skill advancement
29.4	Advanced CTFs	20	Hack The Box Pro Labs, CTFtime, DEF CON CTF qualifiers	Expert competition
29.5	Bug Bounty Methodology	4	Recon, target selection, vulnerability finding, reporting, patience	Real-world practice
29.6	Responsible Disclosure	2	Process, legal, timelines, communication, bounties	Ethical hacking
29.7	Final Penetration Test	20	Full-scope engagement on personal lab	Capstone demonstration
29.8	Final Portfolio	20	GitHub organization, website, resume, LinkedIn, certifications	Professional presentation
29.9	Final Presentation	10	Record presentation, practice delivery, prepare for interviews	Interview readiness

Labs (Capstone — 24 total)

1. Build and configure secure virtual lab (all previous VMs)
2. Perform comprehensive reconnaissance and enumeration

3. Assess web applications (DVWA, Juice Shop, custom apps)
4. Assess network services (Metasploitable2, Windows)
5. Analyze traffic and logs throughout engagement
6. Secure Linux and Windows systems post-exploitation
7. Assess cloud environment (AWS free tier)
8. Secure Docker container and Kubernetes deployment
9. Investigate events using SIEM (Splunk or ELK)
10. Write automation scripts (Bash, PowerShell, Python)
11. Produce professional penetration testing report
12. Create executive summary
13. Create technical findings with evidence
14. Build risk matrix and business impact analysis
15. Publish complete project on GitHub
16. Build portfolio website
17. Record final presentation (15-20 minutes)
18. Practice technical interview with peer
19. Practice behavioral interview with peer
20. **Perform AI red team assessment on LLM system**
21. **Execute purple team exercise with detection validation**
22. **Complete supply chain security audit**
23. **Submit bug bounty report (real or practice)**
24. Apply to 10+ jobs or freelance projects

Final Deliverables

- ☐ **Executive Summary** (1-2 pages, non-technical)
- ☐ **Technical Penetration Testing Report** (30+ pages)
- ☐ **AI Security Assessment Report** (LLM red team)
- ☐ **Purple Team Exercise Report**
- ☐ **Supply Chain Security Audit Report**
- ☐ **Risk Ratings & Business Impact Analysis**
- ☐ **Screenshots & Evidence Collection** (annotated, 100+)
- ☐ **Network Diagrams** (before and after, professional)
- ☐ **GitHub Repository** (complete academy, organized, 25+ repos)

- ☐ **Portfolio Website** (GitHub Pages, live, professional)
- ☐ **Presentation Slides** (15-20 slides)
- ☐ **Presentation Recording** (15-20 minutes)
- ☐ **Lessons Learned Document**
- ☐ **Job Applications** (10+ submitted)
- ☐ **Certification Study Plan** (next 3 certs)
- ☐ **Published Research/Blog Post**

☐ **CERTIFICATION MASTER PLAN {#certifications}**

Tier 1: Foundation (Complete in First 16 Weeks)

Certification	When	Cost	Value	Why Get It
CompTIA Security+	Week 12-16	\$404	☆☆☆☆☆	HR filter, DoD 8570, foundation
CompTIA Network+	Week 8-12	\$358	☆☆☆☆	Networking validation
CompTIA Linux+	Week 12-16	\$358	☆☆☆☆	Linux validation
AWS Cloud Practitioner	Week 14-18	\$100	☆☆☆	Cloud basics
Microsoft AZ-900	Week 14-18	\$99	☆☆☆	Azure basics
LPI Linux Essentials	Week 8-12	\$120	☆☆☆	Linux foundation

Tier 2: Practical Pentesting (Complete in Weeks 17-40)

Certification	When	Cost	Value	Why Get It
eJPT	Week 20-24	\$200	☆☆☆☆	Practical pentesting, great stepping stone
PNPT	Week 28-32	\$300	☆☆☆☆☆	Practical network pentest, TCM reputation
OSCP	Week 36-48	\$1,499+	☆☆☆☆☆	Industry gold standard, hardest
Burp Suite Certified	Week 30-34	\$99	☆☆☆☆	Web security specialization
eWPT	Week 30-34	\$400	☆☆☆☆	Web pentesting

Certification	When	Cost	Value	Why Get It
				practical
CompTIA PenTest+	Week 28-32	\$404	☆☆☆	Pentesting theory + practice
CRTP	Week 36-40	\$300	☆☆☆☆	AD attacks practical

Tier 3: Specialization (Complete in Weeks 40-64+)

Certification	When	Cost	Value	Why Get It
OSWE	Week 40-52	\$1,499+	☆☆☆☆☆	Web app exploitation expert
OSEP	Week 44-56	\$1,499+	☆☆☆☆☆	Advanced pentesting, evasion
OSED	Week 48-60	\$1,499+	☆☆☆☆	Windows exploit development
AWS Security Specialty	Week 44-52	\$300	☆☆☆☆	Cloud security deep dive
Azure Security Engineer (AZ-500)	Week 44-52	\$165	☆☆☆☆	Azure security
CKS	Week 48-56	\$395	☆☆☆☆	Kubernetes security
GCIH	Week 48-60	\$7,000+	☆☆☆☆☆	SANS incident handling
GCIA	Week 52-60	\$7,000+	☆☆☆☆☆	SANS IDS/IPS analysis
GREM	Week 52-60	\$7,000+	☆☆☆☆☆	SANS reverse engineering
CISSP	Week 52-64	\$749	☆☆☆☆☆	Management, 5 years exp required
OSCE3	Post-Academy	\$4,500+	☆☆☆☆☆	Expert-level (OSCP+OSEP+OSSED)

Tier 4: AI Security (Post-Master's or Parallel)

Certification	When	Cost	Value	Why Get It
AWS ML Specialty	Post-Academy	\$300	☆☆☆☆	ML on AWS
Google	Post-Academy	\$200	☆☆☆☆	ML engineering

Certification	When	Cost	Value	Why Get It
Professional ML Engineer				
TensorFlow Developer	Post-Academy	\$100	☆☆☆	ML framework

Certification Study Strategy

Week 1-16: Security+ (primary) + Network+ (secondary)
 Week 17-24: eJPT (primary) + Linux+ (if not done)
 Week 25-32: PNPT (primary) + Burp Suite Certified
 Week 33-40: OSCP preparation (lab time, exercises)
 Week 41-48: OSCP exam + CRTP
 Week 49-56: OSWE or OSEP (choose specialization)
 Week 57-64: CKS or GCIH + CISSP (if eligible)
 Post-Academy: OSCE3, GREM, or AI Security certs

☐ CTF PROGRESSION PATH {#ctfs}

Phase 1: Beginner (Weeks 1-16)

Platform	Path/Room	Skills	Estimated Time
TryHackMe	"Pre-Security"	Basics	1 week
TryHackMe	"Introduction to Cyber Security"	Overview	1 week
TryHackMe	"Linux Fundamentals"	Linux	1 week
TryHackMe	"Network Fundamentals"	Networking	1 week
TryHackMe	"Windows Fundamentals"	Windows	1 week
TryHackMe	"Nmap"	Scanning	3 days
TryHackMe	"OSINT"	Reconnaissance	3 days
PicoCTF	"General Skills"	Linux, scripting	2 weeks
OverTheWire	"Bandit" (all levels)	Linux	2 weeks
CyberDefenders	"Blue Team Labs" (beginner)	Defense	2 weeks

Phase 2: Intermediate (Weeks 17-32)

Platform	Path/Room	Skills	Estimated Time
TryHackMe	"Junior Penetration Tester"	Pentesting	4 weeks
TryHackMe	"Web Fundamentals"	Web security	2 weeks

Platform	Path/Room	Skills	Estimated Time
TryHackMe	"OWASP Top 10"	Web vulnerabilities	2 weeks
Hack The Box	"Starting Point"	Full exploitation	2 weeks
Hack The Box	Easy machines (10+)	Exploitation	4 weeks
PortSwigger Academy	All topics	Web security	4 weeks
OverTheWire	"Natas"	Web + scripting	2 weeks
Root-Me	"Apprentice" challenges	Various	4 weeks
PicoCTF	"Web Exploitation"	Web	2 weeks
CyberDefenders	"Blue Team Labs" (intermediate)	Defense	2 weeks

Phase 3: Advanced (Weeks 33-48)

Platform	Path/Room	Skills	Estimated Time
Hack The Box	Medium machines (20+)	Advanced exploitation	6 weeks
Hack The Box	"Pro Labs" (Dante, CyberNet)	Full pentest	4 weeks
TryHackMe	"Red Team" Path	Red team	2 weeks
TryHackMe	"Active Directory"	AD pentesting	2 weeks
Proving Grounds	Play/Practice	OSCP-style	4 weeks
VulnHub	Various VMs	Full exploitation	4 weeks
Pwnable.kr	Various	Binary exploitation	4 weeks
PicoCTF	"Binary Exploitation"	Memory	2 weeks
CloudGoat	AWS scenarios	Cloud security	2 weeks
PwnedLabs	Various	Practical pentesting	4 weeks

Phase 4: Expert (Weeks 49-64+)

Platform	Path/Room	Skills	Estimated Time
Hack The Box	Hard machines	Expert exploitation	Ongoing
Hack The Box	"Pro Labs" (RastaLabs, CyberNet)	Advanced red team	4 weeks
CTFtime	Live CTFs	Real-time competition	Ongoing
HackerOne	Bug bounty programs	Real-world testing	Ongoing
Bugcrowd	Bug bounty programs	Real-world testing	Ongoing
Intigriti	Monthly challenges	Advanced web	Ongoing

Platform	Path/Room	Skills	Estimated Time
Pwnable.tw	Various	Binary exploitation	Ongoing
OverTheWire	"Behemoth"	Advanced exploitation	Ongoing
SANS NetWars	Various	Defense competitions	Events
DEF CON CTF	Various	World-class competition	Annual
AI Village CTF	AI security challenges	Adversarial ML	Events

CTF Competition Calendar (Major Events)

Event	When	Type	Level
picoCTF	March	Jeopardy	Beginner-Intermediate
BSides CTFs	Various	Jeopardy	All levels
DEF CON CTF	August	Attack-Defense	Expert
HITB CTF	Various	Jeopardy	Intermediate-Advanced
PlaidCTF	April	Jeopardy	Advanced
Google CTF	June	Jeopardy	Advanced
Hack The Box CTF	Various	Jeopardy	Intermediate-Advanced
CyberPatriot	School year	Defense	Beginner-Intermediate
AI Village @ DEF CON	August	AI Security	Intermediate-Advanced

☐ FREELANCE & CAREER ROADMAP {#career}

Freelance Progression

Level 1: Beginner (Weeks 1-16)

Platform	Services	Rate	Weekly Earning Potential
Fiverr	Basic scripts, VM setup, GitHub help	\$10-50	\$50-200
Upwork	IT support, basic troubleshooting	\$15-30/hr	\$300-600
Freelancer	Data entry, basic admin	\$10-25/hr	\$200-500
Direct	Friends/family tech support	\$20-50	Variable

Level 2: Intermediate (Weeks 17-32)

Platform	Services	Rate	Weekly Earning Potential
Fiverr	Security audits, scanner setup, hardening	\$50-300	\$200-1,000
Upwork	Vulnerability assessment, network audit	\$30-60/hr	\$600-1,200
Freelancer	Web security review, log analysis	\$25-50/hr	\$500-1,000
Direct	Small business security consulting	\$50-150/hr	\$500-2,000
HackerOne	Bug bounty (first findings)	\$100-1,000	Variable

Level 3: Advanced (Weeks 33-48)

Platform	Services	Rate	Weekly Earning Potential
Upwork	Pentesting, cloud security, SIEM	\$50-150/hr	\$1,000-3,000
Fiverr	Professional reports, tool development	\$100-500	\$500-2,000
Direct	SMB pentesting, compliance consulting	\$100-300/hr	\$1,000-5,000
HackerOne/ Bugcrowd	Bug bounty (regular findings)	\$500-10,000	Variable
TCM Security	Network pentesting (subcontract)	\$500-2,000/test	Variable

Level 4: Expert (Weeks 49-64+)

Platform	Services	Rate	Weekly Earning Potential
Direct	Enterprise consulting, red team	\$200-500/hr	\$2,000-10,000
Upwork	Expert-level projects	\$100-300/hr	\$2,000-6,000
HackerOne/ Bugcrowd	Bug bounty (critical findings)	\$1,000-50,000+	Variable
Consulting Firms	Subcontract pentesting	\$1,000-5,000/test	Variable
Training	Security training delivery	\$100-300/hr	\$2,000-6,000

Platform	Services	Rate	Weekly Earning Potential
Speaking	Conference talks, webinars	\$500-5,000/talk	Variable
AI Security Consulting	LLM red team, AI safety	\$200-500/hr	\$2,000-10,000

Job Progression Path

Path A: Offensive Security (Red Team)

Year 1: Junior Penetration Tester (\$75k-90k)
 Year 2: Penetration Tester (\$90k-110k)
 Year 3: Senior Penetration Tester (\$110k-140k)
 Year 4: Red Team Operator (\$120k-160k)
 Year 5: Senior Red Team (\$140k-180k)
 Year 6+: Principal Red Team / Consultant (\$160k-250k+)

Path B: Defensive Security (Blue Team)

Year 1: SOC Analyst Tier 1 (\$55k-70k)
 Year 2: SOC Analyst Tier 2 (\$65k-80k)
 Year 3: SOC Analyst Tier 3 / Lead (\$75k-95k)
 Year 4: Threat Hunter (\$85k-110k)
 Year 5: Detection Engineer (\$95k-125k)
 Year 6+: SOC Manager / CISO (Junior) (\$120k-180k+)

Path C: Security Engineering

Year 1: Junior Security Engineer (\$75k-90k)
 Year 2: Security Engineer (\$90k-110k)
 Year 3: Senior Security Engineer (\$110k-140k)
 Year 4: Staff Security Engineer (\$130k-170k)
 Year 5: Principal Security Engineer (\$150k-200k)
 Year 6+: Security Architect (\$170k-250k+)

Path D: Application Security

Year 1: Application Security Analyst (\$75k-90k)
 Year 2: Application Security Engineer (\$90k-115k)
 Year 3: Senior AppSec Engineer (\$110k-140k)
 Year 4: Staff AppSec Engineer (\$130k-170k)
 Year 5: Principal AppSec Engineer (\$150k-200k)
 Year 6+: AppSec Architect / Director (\$170k-250k+)

Path E: Cloud Security

Year 1: Junior Cloud Security Engineer (\$85k-100k)
 Year 2: Cloud Security Engineer (\$100k-125k)
 Year 3: Senior Cloud Security Engineer (\$125k-160k)
 Year 4: Cloud Security Architect (\$150k-200k)
 Year 5: Principal Cloud Security (\$170k-220k)
 Year 6+: Cloud Security Director (\$200k-300k+)

Path F: AI Security (Your Specialization)

Year 1:	AI Security Analyst	(\$90k-110k)
Year 2:	AI Security Engineer	(\$110k-140k)
Year 3:	Senior AI Security Engineer	(\$140k-180k)
Year 4:	AI Red Team Lead	(\$160k-220k)
Year 5:	Principal AI Security	(\$180k-250k)
Year 6+:	AI Security Architect / Director (\$200k-350k+)	

Path G: Freelance / Independent

Year 1:	Part-time freelancer	(\$10k-30k side income)
Year 2:	Full-time freelancer	(\$50k-100k)
Year 3:	Established consultant	(\$100k-200k)
Year 4:	Security consultancy owner	(\$200k-500k)
Year 5+:	Boutique firm / agency	(\$500k-2M+)

Salary Benchmarks by Region (US, 2026)

Role	Entry	Mid	Senior	Principal
SOC Analyst	\$55k-70k	\$70k-90k	\$90k-120k	\$120k-150k
Penetration Tester	\$75k-90k	\$90k-120k	\$120k-160k	\$160k-220k
Red Team	\$90k-110k	\$110k-140k	\$140k-180k	\$180k-250k
Security Engineer	\$80k-100k	\$100k-130k	\$130k-170k	\$170k-230k
Cloud Security	\$90k-110k	\$110k-140k	\$140k-180k	\$180k-250k
AppSec Engineer	\$85k-105k	\$105k-135k	\$135k-175k	\$175k-240k
Security Consultant	\$85k-110k	\$110k-150k	\$150k-200k	\$200k-300k
Security Architect	\$120k-150k	\$150k-200k	\$200k-280k	\$280k-400k
AI Security Engineer	\$90k-110k	\$110k-150k	\$150k-200k	\$200k-300k
AI Red Team Lead	\$120k-150k	\$150k-200k	\$200k-280k	\$280k-400k
CISO (Junior)	\$150k-200k	\$200k-280k	\$280k-400k	\$400k-600k
Bug Bounty Hunter	Variable	Variable	Variable	Variable
	\$10k-50k	\$50k-150k	\$150k-500k	\$500k-1M+

☐ DAILY SCHEDULE TEMPLATE {#schedule}

Standard Study Day (4 Hours)

Time	Activity	Details
0:00-0:30	Review & Quiz	Review previous day, quick quiz
0:30-1:30	Theory Study	Read, watch videos, take notes
1:30-2:00	Break	Step away, hydrate, stretch
2:00-3:30	Hands-On Lab	Practice, experiment, break things
3:30-4:00	Documentation	Update GitHub, write notes, screenshots

Weekend Deep Dive (6-8 Hours)

Time	Activity	Details
0:00-1:00	Week Review	Review week's learnings
1:00-2:00	CTF Practice	TryHackMe, Hack The Box, etc.
2:00-3:00	Break	Rest, exercise, social
3:00-5:00	Build Project	Work on current build week project
5:00-6:00	GitHub Update	Polish documentation, push code
6:00-8:00	Certification Study	Practice exams, review

Build Week Schedule (40 Hours)

Day	Focus	Hours
Monday	Planning & Design	8
Tuesday	Development/Implementation	8
Wednesday	Development/Testing	8
Thursday	Documentation & Polish	8
Friday	Review & Presentation	8

🔧 TOOLS & RESOURCES MASTER LIST {#tools}

Virtualization

- **VirtualBox** (free) — Primary hypervisor
- **VMware Workstation Pro** (paid) — Alternative
- **Vagrant** (free) — Automated VM deployment
- **Docker** (free) — Containerization
- **Minikube/kind** (free) — Kubernetes local

Target Machines

- **Metasploitable2/3** — Vulnerable Linux/Windows
- **DVWA** — Web app vulnerabilities
- **WebGoat** — Web security
- **Juice Shop** — Modern web app
- **VulnHub** — Various vulnerable VMs
- **Hack The Box** — Live targets
- **TryHackMe** — Guided targets
- **CloudGoat** — AWS exploitation scenarios
- **Kubernetes Goat** — K8s exploitation
- **AI Village CTF** — AI security challenges

Scanning & Enumeration

- **Nmap** — Network scanner
- **Masscan** — High-speed scanner
- **RustScan** — Modern fast scanner
- **Nessus** — Vulnerability scanner
- **OpenVAS** — Open-source vulnerability scanner
- **Nuclei** — Fast vulnerability scanner
- **AutoRecon** — Automated enumeration
- **Legion** — Automated enumeration

Web Security

- **Burp Suite** (Community/Pro) — Web proxy
- **OWASP ZAP** — Free web proxy
- **SQLMap** — SQL injection automation

- **Nikto** — Web scanner
- **Gobuster** — Directory brute force
- **Ffuf** — Fast web fuzzer
- **Wappalyzer** — Technology detection

Exploitation

- **Metasploit Framework** — Exploitation framework
- **Cobalt Strike** (paid) — Advanced red team
- **Impacket** — Python network protocols
- **Mimikatz** — Credential extraction
- **BloodHound** — AD analysis
- **PowerSploit** — PowerShell exploitation
- **Empire** — Post-exploitation

Password Security

- **Hashcat** — GPU hash cracking
- **John the Ripper** — CPU hash cracking
- **Hydra** — Online password cracking
- **CrackStation** — Online hash lookup
- **CeWL** — Custom wordlist generator

Active Directory

- **BloodHound** — AD attack path analysis
- **SharpHound** — BloodHound data collector
- **PowerView** — AD reconnaissance
- **Rubeus** — Kerberos abuse
- **Certipy** — AD CS exploitation

Wireless

- **Aircrack-ng** — Wireless suite
- **Wireshark** — Packet analyzer
- **Kismet** — Wireless detector
- **WiFi Analyzer** — Android WiFi tool

Mobile

- **Android Studio** — Android emulator
- **apktool** — APK decompilation
- **jadx** — APK decompiler
- **MobSF** — Mobile security framework
- **Frida** — Dynamic instrumentation
- **Objection** — Mobile runtime exploration

Cloud

- **AWS CLI** — AWS command line
- **Azure CLI** — Azure command line
- **gcloud** — GCP command line
- **ScoutSuite** — Cloud security audit
- **Prowler** — AWS security audit
- **CloudSploit** — Cloud security scanner
- **Steampipe** — Cloud SQL queries

Containers

- **Docker** — Container platform
- **Trivy** — Container scanner
- **Clair** — Container scanner
- **Snyk** — Container + dependency scanner
- **Falco** — Runtime security
- **kube-bench** — K8s security checker

SIEM & Monitoring

- **Splunk** (free trial) — SIEM
- **ELK Stack** (free) — Open-source SIEM
- **Wazuh** — Open-source security monitoring
- **OSSEC** — Host-based IDS
- **Snort** — Network IDS
- **Suricata** — Network IDS/IPS
- **Sigma** — Generic detection rules
- **YARA** — Pattern matching for malware

Memory & Binary

- **GDB** — Debugger
- **pwndbg** — GDB enhancement
- **pwntools** — Python exploit development
- **Radare2** — Reverse engineering
- **Ghidra** — Reverse engineering (free)
- **IDA Free** — Reverse engineering
- **x64dbg** — Windows debugger

AI Security

- **PyTorch** — ML framework
- **TensorFlow** — ML framework
- **Jupyter** — Interactive notebooks
- **ART (Adversarial Robustness Toolbox)** — Adversarial ML
- **Foolbox** — Adversarial attacks
- **TextAttack** — NLP adversarial attacks
- **Hugging Face Transformers** — LLM security testing

Purple Team

- **Atomic Red Team** — MITRE ATT&CK atomic tests
- **CALDERA** — Adversary emulation
- **Prelude Operator** — Continuous security testing
- **Vectra** — AI-driven threat detection

IaC & DevSecOps

- **Terraform** — Infrastructure as code
- **tfsec** — Terraform security scanner
- **checkov** — IaC security scanner
- **GitHub Actions** — CI/CD
- **GitLab CI** — CI/CD
- **Cosign** — Container signing
- **Syft** — SBOM generation
- **Vault** — Secret management

Documentation & Productivity

- **GitHub** — Version control, portfolio
- **GitHub Pages** — Free website hosting
- **Obsidian** — Knowledge management
- **Notion** — Project management
- **draw.io** — Diagrams
- **Markdown** — Documentation
- **LaTeX** — Professional reports

Learning Resources

- **TryHackMe** — Guided learning
- **Hack The Box** — Practice labs
- **PortSwigger Academy** — Web security
- **PicoCTF** — Beginner CTFs
- **OverTheWire** — Linux wargames
- **CyberDefenders** — Blue team
- **BTLO** — Blue team
- **SANS** — Premium training
- **Offensive Security** — Premium training
- **TCM Security** — Affordable training
- **eLearnSecurity** — Practical training
- **AI Village** — AI security
- **MITRE ATT&CK** — Threat framework

☐ WHAT YOU'LL BUILD (COMPLETE INVENTORY)

By the end of this academy, you will have:

Labs & Infrastructure

- Complete virtual cybersecurity lab (9+ VMs)
- Cloud environment (AWS/Azure/GCP)
- Container environment (Docker + Kubernetes)
- SIEM environment (Splunk + ELK)
- Wireless testing setup

- Mobile testing environment
- Active Directory lab domain
- AI/ML testing environment

Scripts & Tools

- Linux Administration Toolkit (15+ Bash scripts)
- Windows Hardening Toolkit (15+ PowerShell scripts)
- Python Security Toolkit (20+ scripts)
- Custom network scanner
- Custom vulnerability scanner
- Log analysis automation
- Incident response automation
- Report generation automation
- AI security testing toolkit
- Detection engineering toolkit

Reports & Documentation

- Network Analysis Report
- Vulnerability Assessment Report
- Web Application Penetration Test Report
- Wireless Security Audit Report
- Mobile Security Assessment Report
- Cloud Security Assessment Report
- Container Security Report
- SIEM Investigation Report
- AI Security Assessment Report
- Purple Team Exercise Report
- Supply Chain Security Audit Report
- Malware Analysis Report
- Final Capstone Penetration Test Report
- Professional Report Templates

Portfolio & Professional

- Complete GitHub portfolio (organized, documented, 25+ repos)

- Personal portfolio website (GitHub Pages)
- Professional resume (cybersecurity-focused)
- Optimized LinkedIn profile
- CTF profiles (TryHackMe, Hack The Box, etc.)
- Bug bounty profiles (HackerOne, Bugcrowd)
- Presentation recording (15-20 minutes)
- Interview preparation (50+ questions)
- Published blog post or research
- Open source contribution

Certifications (Target)

- CompTIA Security+
- eJPT or PNPT
- OSCP (or in progress)
- Cloud certification (AWS/Azure/GCP)
- Specialized certification (OSWE/OSEP/CKS)
- AI/ML certification (post-Master's)

III SUCCESS METRICS

Metric	Target	Measurement
GitHub Repositories	25+	Count of organized repos
Scripts Written	50+	Count of original scripts
CTF Challenges Solved	150+	TryHackMe + HTB + others
Machines Compromised	75+	HTB + VulnHub + lab
Reports Written	20+	Professional reports
Certifications Earned	5-7	Security+ + practical + cloud + AI
Blog Posts/Writeups	20+	Published writeups
Bug Bounty Reports	5+	Valid submissions
Job Applications	50+	Applications sent
Interviews Completed	10+	Technical + behavioral
Freelance Projects	5+	Completed projects
LinkedIn Connections	500+	Security professionals
Conference Talks	1+	Local or virtual

Metric	Target	Measurement
Open Source Contributions	3+	Merged PRs
AI Security Projects	3+	Adversarial ML demos

☐ QUICK START CHECKLIST

Before Week 1

- ☐ Computer with 16GB+ RAM (32GB preferred)
- ☐ 500GB+ free disk space (SSD preferred)
- ☐ Reliable internet connection
- ☐ GitHub account created
- ☐ LinkedIn account created
- ☐ TryHackMe account created
- ☐ Hack The Box account created
- ☐ VirtualBox downloaded and installed
- ☐ Kali Linux ISO downloaded
- ☐ Windows ISO downloaded
- ☐ Metasploitable2 downloaded
- ☐ Study space organized (quiet, comfortable)
- ☐ Calendar blocked for study time (4 hrs/day)
- ☐ Family/roommates informed of study schedule
- ☐ Notebook or digital note-taking system ready
- ☐ Motivation: Write down WHY you're doing this
- ☐ **Set up Python environment with PyTorch/TensorFlow**
- ☐ **Install Jupyter Lab for AI experiments**

Week 1 Day 1

- ☐ Install VirtualBox
- ☐ Create first VM
- ☐ Push first README to GitHub
- ☐ Join cybersecurity Discord/Slack communities
- ☐ Follow 10+ security professionals on Twitter/X

- ☐ Subscribe to 5+ security YouTube channels
- ☐ Set up study playlist (focus music)
- ☐ Take "before" photo of your study space
- ☐ Write commitment statement
- ☐ Start Day 1 of the academy
- ☐ **Set up AI/ML environment (PyTorch, Jupyter)**

YEAR 2 SPECIALIZATION TRACKS {#year2}

After completing the 60-week core academy, choose your specialization:

Track 1: AI Security Engineer (Your Track)

Duration: 6 months

Focus: Adversarial ML, secure AI deployment, AI red teaming

Module	Duration	Topics
Advanced Adversarial ML	6 weeks	Advanced attacks, defenses, certified robustness
Secure MLOps	4 weeks	Model serving, monitoring, A/B testing security
AI Governance & Risk	4 weeks	NIST AI RMF, EU AI Act, model cards, bias
AI Red Team Operations	4 weeks	LLM red team, model extraction, supply chain
Research Project	6 weeks	Publish paper or CVE in AI security

Target Certifications: AWS ML Specialty, Google Professional ML Engineer

Target Roles: AI Security Engineer, AI Red Team Lead, ML Security Researcher

Target Salary: \$150k-\$300k+

Track 2: Red Team Operator

Duration: 6 months

Focus: Advanced evasion, custom tooling, adversary emulation

Module	Duration	Topics
Advanced Evasion	6 weeks	EDR bypass, custom implants, C2 development
Physical & Social Engineering	4 weeks	Badge cloning, tailgating, vishing, pretexting
Custom Tool Development	4 weeks	C2 frameworks, implants, post-

Module	Duration	Topics
		exploitation tools
Adversary Emulation	4 weeks	APT emulation, TTP mapping, purple team
Research Project	6 weeks	Publish research, tool, or CVE

Target Certifications: OSEP, OSCE3

Target Roles: Senior Red Team, Principal Red Team, Offensive Security Researcher

Target Salary: \$160k-\$280k+

Track 3: Blue Team Lead

Duration: 6 months

Focus: Detection engineering, threat hunting, SOC leadership

Module	Duration	Topics
Advanced Detection Engineering	6 weeks	Custom detection, ML-based detection, behavioral analytics
Threat Hunting Mastery	4 weeks	Hypothesis-driven hunting, threat intel integration
SOC Leadership	4 weeks	Team management, metrics, incident response leadership
Digital Forensics Deep Dive	4 weeks	Memory forensics, disk forensics, malware reverse engineering
Research Project	6 weeks	Publish detection research or threat intel

Target Certifications: GCIA, GCFA, GCTI

Target Roles: SOC Manager, Detection Engineering Lead, Threat Hunting Lead

Target Salary: \$140k-\$250k+

Track 4: Malware Analyst / Reverse Engineer

Duration: 6 months

Focus: Deep binary analysis, malware reverse engineering, exploit development

Module	Duration	Topics
Advanced Reverse Engineering	6 weeks	IDA Pro, Ghidra scripting, deobfuscation
Malware Analysis Mastery	4 weeks	Ransomware, APT malware, banking trojans
Windows Internals	4 weeks	Kernel, drivers, rootkits
Exploit Development	4 weeks	Browser exploits, kernel exploits, 0-day research

Module	Duration	Topics
Research Project	6 weeks	Publish CVE or malware analysis

Target Certifications: GREM, OSED, OSCE3

Target Roles: Senior Malware Analyst, Principal Reverse Engineer, Exploit Developer

Target Salary: \$160k-\$300k+

Track 5: Security Architect

Duration: 6 months

Focus: Enterprise security design, zero trust, cloud architecture

Module	Duration	Topics
Enterprise Security Architecture	6 weeks	Zero trust, SASE, security architecture patterns
Cloud Security Architecture	4 weeks	Multi-cloud security, cloud-native security
Security Governance	4 weeks	Risk management, compliance, GRC
Emerging Technologies	4 weeks	Quantum security, blockchain security, IoT security
Research Project	6 weeks	Design and publish security architecture

Target Certifications: CISSP, CCSP, SABSA

Target Roles: Security Architect, Principal Security Architect, CISO (Junior)

Target Salary: \$180k-\$350k+

□ FINAL VERDICT

This is the most comprehensive, elite cybersecurity academy roadmap available.

It covers:

- All foundational skills (networking, Linux, Windows)
- Active Directory deep dive (enterprise backbone)
- Complete offensive security (recon, scanning, enumeration, exploitation)
- Complete defensive security (SIEM, detection engineering, threat hunting)
- Modern web security (including LLM attacks, GraphQL, APIs)
- Cloud security (AWS, Azure, GCP)
- Container security (Docker, Kubernetes)

- AI Security & Adversarial ML (your Master's specialization)
- Reverse engineering & malware analysis
- Purple team operations
- Infrastructure as code & supply chain security
- Professional skills & career development

Duration: 72 weeks (~18 months) at 20 hrs/week = 1,440+ hours

Output: Elite cybersecurity professional ready for any role including AI Security

In shā' Allāh, this is the academy we'll complete together.

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